

**Developing a Novel Carbon Offset Program in the University of Southern
California's Surrounding Communities of South and East Los Angeles**

A Research Report Developed for USC's Presidential Working Group on
Sustainability

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Executive Summary

This report provides recommendations for carbon offset projects, as well as recommendations for necessary further research, that are the result of a research project conducted by USC Sol Price School of Public Policy graduate students on behalf of the USC Presidential Working Group on Sustainability and USC President Dr. Carol Folt. The motivation for this research project is to explore the development of a novel hyper-local carbon offset strategy for the University of Southern California. Such a program would facilitate the reduction of USC's carbon emissions deficit by investing in energy efficiency, behavioral change, and other carbon-reducing measures within the local communities surrounding the campuses of the University. The recommendations in this report are made in service of the University's ambitious goal to reach carbon neutrality by 2028, taking into account specific priorities of the Presidential Working Group on Sustainability to improve the health, wellness, and economic outlook of local communities via carbon offset purchases and investments made on a local scale.

There is no centralized market for local carbon offsets in USC's surrounding communities, and there is no concrete mechanism in place for the University to purchase carbon offsets from other local entities. The research team proposes a novel framework for how local carbon offsets might function in a practical application by the University, an innovation that would be a first-of-its-kind carbon offset strategy. The researchers worked with local stakeholders and experts from around the United States in fields ranging from psychology to civil engineering in order to develop the recommendations that are outlined in this report.

A hyper-local carbon offset strategy would have political, budgetary, and other implications, including direct impacts on the community as a result of University actions. Therefore, the research team sought to understand the theoretical frameworks surrounding carbon offsets, co-impacts of carbon emissions reduction projects, and carbon markets as a means to design an effective carbon offset strategy that would engender both university and community support. In some cases, this meant looking at what other universities have done to create an impact in their local communities. In other cases, it meant reimagining the role of USC as a local actor, thought leader, and developer of both public and private infrastructure within South and East Los Angeles. As the largest private employer in Los Angeles, USC's commitment to the green economy offers opportunities for job growth, creating citywide change, and a positive impact on businesses and residents of the communities of South and East Los Angeles. The technical

innovations that USC advances in its quest for carbon neutrality could advance sustainability in the community and leave residents with the knowledge, tools, and desire to live increasingly carbon-neutral lifestyles.

This report uses a host of analytical and research methods. The primary research questions that guided our methodology were:

1. What are the criteria used to evaluate offsets and carbon-reduction projects? What types of projects are cost-effective ways to offset carbon locally? What projects are most effective in reducing carbon emissions?
2. What are urban areas and universities doing to offset carbon (i.e., types of projects or technologies)? What are their impacts?
3. Based on community feedback and expert perspectives, how can USC's offset plan best appeal to the neighborhoods surrounding campus? What opportunities for partnerships exist?

As such, the selected methods were meant to both inform the answers to and assist in the contextualization of these questions. Our research methods included a literature review, expert interviews, and stakeholder engagement. We identified concepts and trends in the existing literature and used that information to guide further research and our discussions with experts we identified. Our regular meetings with the client also helped to guide our outreach and our interview protocol with experts, who ranged from University faculty and staff to behavioral scientists and university sustainability teams from other parts of the country. In addition, we utilized our access to USC personnel engaged with the Los Angeles community to deepen our understanding of University-community relationships. Finally, we were able to host a community listening session with leaders and resident members of the communities around USC's campuses, where we discussed our project with and heard perspectives from those individuals. Altogether, our research benefited greatly from the access to these diverse stakeholders.

In terms of analytical methods, we employed criteria-alternative matrices to compare the various alternatives. For alternatives, we used financial modeling, cost-benefit analysis, and sensitivity analysis to produce estimates of the net present cost to then compare against a baseline cost of \$1.38 million to offset USC's Scope 1, Scope 2, and Scope 3 emissions (92,000 metric tons CO₂). This baseline represents the cost of offsetting 92,000 metric tons of CO₂ on the international market, with a price of \$15/metric ton. As all alternatives will reduce an equivalent amount of CO₂ emissions, we used net present cost instead of net present value because we are comparing the alternatives based on monetary cost, not monetized benefits. Yet, this is nonetheless a cost-benefit analysis, not a cost-effectiveness analysis due to framing, as we are comparing the outright costs and separately listing the co-benefits of alternatives. Then, once the estimates were generated, they were evaluated across two sets of criteria: the first set of criteria is a result of initial research and includes: Cost, Feasibility/Efficiency, Co-Benefits to

Community, Emissions Reduction Potential, and Equity. The second set of criteria uses an industry standard referred to as P.A.V.E.R., which stands for Permanent, Additional, Verifiable, Enforceable, and Real (California Health and Safety Code §38562(d)(1) and (d)(2)). We constructed two criteria-alternatives matrices, one with each set of criteria. Then, for the first matrix, using the research and outcomes of our analyses, each alternative is given a score for each criterion between “1” and “5,” with scores signaling ascending favorability. We then total the scores for each, with any score greater than or equal to “12” indicating that the alternative is a potentially viable option. For the P.A.V.E.R. matrix, each alternative is evaluated across each criterion on a binary scale, with “1” meaning it meets the criterion and “2” meaning it does not.

The strengths of this method are numerous and have broad applicability. Because this report evaluates novel ideas for local carbon offsets projects, we needed a methodology that could simultaneously inform and compare alternatives that are very different from one another and that are speculative in nature. This research profited from access to key experts and to community stakeholders. Furthermore, quantitative analysis gave us a level of objective footing with which to compare and contrast the alternatives. Finally, the criteria-alternative matrices were well-researched and allow us to compare viable alternatives based on their individual criterion scores, as well.

However, this methodology is not without limitations. Firstly, due to time constraints and the COVID-19 pandemic, there are some alternatives that did not receive attention but that deserve further research. For example, these include some ecological projects, such as urban forestry, or projects for carbon *sequestration* rather than reduction or avoidance. Next, there may be opportunities to generate other new ideas through Request For Proposals (RFPs), which this project was unable to do. In addition, some projects have a degree of risk that was difficult for us to assess. Similarly, our criteria-alternative matrices include a necessary degree of subjectivity. Future examinations of project ideas should aim to reduce this as much as possible. Furthermore, without a cost- or budget constraint, we are unable to give a strict recommendation. Some projects may reap many co-benefits but the quantification of them is limited and different aspects of the projects may also be quite expensive. Finally, while we are able to engage with some community stakeholders, we recommend elevating community voices when considering an investment in these or other alternatives going forward.

The weatherization project primarily entails sponsoring building envelope improvements, appliance upgrades, and all-energy system interventions in low-income single family homes. A project targeting weatherization of homes was previously implemented by Duke University’s Home Energy Loan (HEAL). Studies of weatherization through federal programs have been proven to be effective, leading to 15-20% energy usage reduction on average per household. The downside of this type of offset project is that most home weatherization projects have low sign-up rates and the estimates of their effectiveness is currently being debated.

A residential solar panel project would be another proposed plan that would shift the energy efficiency of a household, but be more costly than weatherization. Like the weatherization project, this project could potentially be less appealing due to its nature as a long-term investment that requires maintenance. Nevertheless, the long-term benefit and energy savings would be immense and solarization could have more distributed benefits if implemented in multi-units or low-income rental facilities, such as those managed by NGOs. Like weatherization, a solar project would entail an upfront investment for households in the USC Good Neighbors Campaign, who may typically face financial burdens. However, because of high-upfront costs these types of projects may be less feasible communities with high proportions of renters.

The small business solarization would be a reduction in the measured carbon footprint of small businesses within the Good Neighbors Campaign. The project proposed in this report describes the potential to reduce 244,000 tons CO₂ per year, exceeding the 92,000 tons CO₂ deficit per year that USC planned to offset. The small businesses solarization project would mainly benefit from long-term energy savings and USC would bear the financial burden of installation and/or maintenance of the solarization devices.

The healthy food market plan involves the investment in local, small grocery stores and the development of partnership between them. The consumers of healthy food from these stores would be the beneficiaries of this program and the shifting dietary paradigms for USC's community members would further reduce their carbon footprint every year. The costs associated with this plan would range from 1.6 million to 2.5 million and the outcome would be difficult to measure because people's dietary intakes are hard to monitor.

The last plan would be an FCEV sharing program that provides environmentally conscious alternatives since transportation comprises a large portion of energy usage in the area. This type of project was implemented in San Bernardino and Riverside and it would only cost \$5 per hour to use the car. The cost is as expensive as public transit but more versatile when people need to commute to areas that may have public safety challenges. The total cost for this program would be approximately \$681,421, which could be costly for a fleet of FCEVs and their associated infrastructure.

All of the proposed options in this report are more costly than purchasing credits from the international carbon offset market. However, USC's surrounding demographics in South and East Los Angeles might derive significant co-benefits. The meeting with community residents from these communities made it clear that leaders, partners, and organizations nearby want to be included in USC's carbon neutrality and potential offset strategy plans.

The expert and community outreach was a strength of the methodology. We conducted interviews with professors and staff from USC, experts from other organizations, and with a panel of community members from the University Park Campus and Health Sciences Campus neighborhoods. The main themes that arose in discussions with USC experts were to use the Good Neighbors apparatus where possible, to consider invoking RFPs to generate ideas and build relationships, to promote and frame research opportunities for faculty and students, to partner with other institutions and nonprofits, and to consider creative ways to engage the student population. Interviews with external experts highlighted the importance of carbon offsets protocols, of considering how USC wants to approach peer review of offsets and reporting offsets, and to emphasize the University's role as a public educator. Finally, our listening session with the panel of community leaders highlighted the importance of blended, normalized efforts between USC and the surrounding communities and highlighted the importance of community-wide rather than individual-level interventions.

The findings of this report primarily indicate that more investigation into carbon offsetting projects must be conducted on behalf of USC. As demonstrated throughout the report, in order to create actual carbon reductions, the projects must meet a number of important criteria and go through a verification process for developing credibility (Lee and Mayer, 2020). It can be difficult for any given project to meet all of the criteria required for offsetting and it will be important for USC to set priorities in terms of criteria the developed projects should meet. Prepared as an initial exploration of carbon offsetting, this report has determined that while the development of offset projects can present numerous challenges and that no single project is without risks (See *Appendix B* for a description of different types of offset risks), the energy savings benefits and co-benefits to local communities may justify offsets that support the local transition to low-carbon energy systems (Lee and Mayer, 2020).

Based on the project research, the report makes the following recommendations to USC's PWG:

1. Continue to explore localized offsets with consideration of projects that would offer widespread benefits in the community, with support and guidance from community members, university faculty, research centers, and students
2. Diversify the university's offsets portfolio with a combination of international offsets and a strategy for localized pilot programs
3. Engage in authentic community engagement and outreach for project development and draw from the university's research & academic expertise to formulate a portfolio of offsets
4. Collaborate with a diverse set of CBOs, partners, and higher education institutions to develop projects going forward and create momentum for climate change mitigation

Issue Diagnosis

An emerging report by the University of Southern California Office of Sustainability highlights that despite operational efforts to reduce greenhouse gas (GHG) emissions outlined in the university's 2028 Sustainability Plan, USC faces an emissions gap of an estimated 50,000 metric tons of CO₂ per year for Scope 1 and 2 emissions and an estimated 92,000 metric tons of CO₂ per year when Scope 3 emissions are taken into account. The university is committed to aggressively pursuing carbon neutrality as an element of the Operations Sustainability 2028 Plan currently under development. Given the university's goal to reach carbon neutrality by 2028, the USC Presidential Working Group on Sustainability (PWG) is developing a carbon offset strategy to help close this significant gap.

This report provides findings from a research project meant to advise the PWG regarding options to create a first-of-its-kind program of community-based carbon offset projects in Los Angeles. The aim of this research project is to advise on the options available to the university to achieve its goal of carbon neutrality by 2028. This ambitious goal will require purchasing or investing in carbon offsets external to university operations. In developing a strategy for these offsets, the PWG has an opportunity to either purchase offset credits from an international market or to engage with the local community in Los Angeles to develop a novel hyper-local carbon offset program.

Overview of Carbon Markets

The PWG's local carbon offset project proposal is groundbreaking because it reimagines how carbon offsets can work in a community-oriented context. Traditionally, carbon offsets are purchased on an international market. Organizations purchase offsets that fund greenhouse gas (GHG) emissions reduction projects in other areas of the world, thus offsetting their own carbon emissions. In this report, the practicum researchers outline a method whereby USC would fund GHG emissions reductions (university carbon offsets) directly in Los Angeles. To further clarify the unique nature of this research, the following sections provide an overview of carbon markets and economic frameworks that justify carbon offsetting on an institutional basis, as well as a dive into the distributional effects of carbon offsetting on a global and local scale.

The first international carbon market system was established by the Kyoto Protocol in 1997 (Environmental Defense Fund, 2020). This market, the Clean Development Mechanism (CDM), allows countries with emissions-reduction commitments to implement emissions-reduction projects in developing countries. Through the CDM, high-emitting countries can offset their emissions through renewable energy investments in other parts of the world. Economists show the CDM has had a "positive impact in terms of cost-effectiveness for renewable energy projects" and the mechanism has catalyzed sustainable policy initiatives around the world (Mele, Paglialunga, & Sforza, 2020). Two additional international carbon markets were established by

the Paris Agreement of 2016, with specific details still in process (Gao et. al., 2019). The proliferation of international carbon markets suggests a global trend toward emissions reduction and sustainability. This trend is supported by a variety of local and regional markets for carbon offsets in geographies around the world.

The market for carbon offsets was valued at over \$144 billion in 2010, with projections expecting a market value of over \$1 trillion by 2020 (Kosoy and Ambrosi, 2010). The value of carbon markets is derived from renewable energy project investments, cap-and-trade program transactions, and other offset investments made by emissions-reducing entities (Fankhauser and Hepburn, 2010). The creation of a market around sustainable emissions reduction is grounded in theoretical frameworks from climate and development economics.

The Economics of Carbon Offsets

Conventional environmental economics establishes the need to price environmental externalities and create market-based solutions to environmental problems (Baumal and Oates, 1988; Kolstad, 2010). Just as public goods provide value to all, public “bads” like carbon emissions cause harm indiscriminately. Establishing a market for the mitigation of negative externalities incentivizes stakeholders to reduce harmful activities – making those activities either too expensive to continue or profitable to mitigate. Environmental market-based solutions have positive impacts that are amplified through economic markets, including increasing revenue, employment, and total factor productivity of downstream firms (Peng et. al., 2020).

Economics further establishes a theory of the spillover effects from carbon offsetting in high-emissions producers. Game theory in particular suggests that as producers purchase carbon offsets in the open market, carbon trading increases the retail price of products from high-emissions producers (Xia et al., 2020). As a result of the relative increase in the price of high-carbon products, markets for low-carbon products from other producers exhibit increased demand. In other words, carbon trading can expedite the transition away from high-carbon products by increasing demand for low-carbon products that become relatively cheaper.

As a market mechanism, carbon offsetting allows businesses, institutions, and individuals to reduce their greenhouse gas emissions (GHGs) by purchasing, trading, or selling a reduction in, or sequestration of, carbon emissions somewhere else, often through credits and across borders (McAfee et al., 2015). This mechanism has received criticism for various reasons. While proponents assert that offsetting increases efficiency gains by facilitating increased GHG reductions at lower costs and greater flexibility, opponents challenge its actual effectiveness in reducing GHGs and the environmental integrity or sustainable quality of the initiatives undertaken (GAO, 2008; McAfee et al., 2015). Some highlight that carbon offsetting has the potential to allow firms to continue to pollute while “purchasing” or “buying” investments in

projects that reduce or sequester carbon, which is often executed by external parties (Anderson and Bernauer, 2016; GAO, 2008; McAfee et al., 2015). By extension, others argue that participation in international markets for carbon offsets disincentivizes businesses and entire sectors from changing source points of pollution, adapting consumption patterns, or making switches to alternative fuels (Cushing et al., 2018; McAfee et al., 2015).

Impacts of Carbon Markets and Offsets on Disadvantaged Communities

Examples of carbon reduction programs and pollution markets have highlighted common issues about ethics, fairness, and social equity. It is important to note that some of the specific features of carbon or GHG pollution and climate change make these problems inherent. For example, GHGs are inherently global, as they are atmospheric gases that do not stay in one place. As significant drivers to global warming, they contribute to a worldwide warming point that can have disastrous consequences for wildlife and humans (McAfee et al., 2015). The uneven distribution of the effects of climate change and the correlation of carbon pollution with increased wealth or income and long-term development of countries and institutions necessarily implicates social justice and fairness. The adverse effects of climate change can transform entire ecosystems with documented impacts on livelihoods and increase the incidence of natural disasters with the potential for imbalanced effects on the poor people and other groups, such as people living in coastal communities (McKinnon, 2015). To date, most American-driven offset projects occur in developing countries, which then have the potential to disrupt livelihoods for rural farmers and the rights of indigenous peoples or proliferate activities that may be unsustainable and environmentally damaging (McAfee et al., 2015).

Focusing solely on climate change's global social justice implications could also undermine local opportunities for improving environmental health and communities. As an example, Cushing et al.'s (2018) empirical study of California's Cap and Trade Program from 2011 to 2015 demonstrates that efforts to mitigate climate change via carbon trading did not lead to increased equity. During the first years of the program, they found that most facilities increased their local GHG emissions despite engaging in significant out-of-state offsetting investments (Cushing et al., 2018). The authors found that exposure to co-pollutant emissions, known to be significantly associated with harmful health outcomes such as cancer and increased mortality, were correlated with high carbon emissions (Nguyen et al., 2018; Cushing et al., 2018). In the study, GHG emitting facilities were close to communities that were higher in density, in their composition of people of color, and in the likelihood of being home to low-income and less educated people (Cushing et al., 2018). Their study highlights how policies that allow polluting facilities to participate in the carbon reduction market as part of their regulatory actions can be problematic for disadvantaged communities located near those facilities (Cushing et al., 2018).

Carbon does not have direct impacts on health at its typical concentrations in the air that humans breathe. However, common co-pollutants of carbon have adverse effects that disproportionately affect disadvantaged communities globally (Nguyen et al., 2018). These co-pollutants, which include particulate matter, nitrous oxides, ozone, and volatile organic compounds, tend to be more prominent in emissions by heavy industries and transportation combustion (Cushing et al., 2018).

USC has the unique opportunity to innovate with an institutional carbon offset strategy that both contributes to the mitigation of climate change and has non-carbon reduction benefits for the community, also known as co-benefits in the literature (van Kooten et al., 2015). Chang et al. (2017) conducted a review of studies published between 2009-2017 and found that there are many sectors that can receive co-benefits from carbon emissions reduction (i.e., building, industry, agriculture). In particular, co-benefits related to other parts of the environment, diet, air quality, and transit are prominent in the literature (Chang et al., 2017). The literature review found that health co-benefits include reductions in disability-adjusted life years and mortality (ibid.). The novelty and diversity of co-benefit models may allow local decision-makers to choose mitigation options that maximize co-benefits that address their specific interests and foci (ibid.).

Co-benefits from Voluntary Offset Projects

USC's interest in underwriting carbon offsets that have local co-benefits is a cutting-edge approach to carbon offset programming. While the traditional approach to carbon offsets is to purchase offsets on the international market, there is increasing acknowledgment of opportunities for local co-benefits. Kountouris et al. (2014) examine voluntary projects supported by corporate buyers internationally, finding that most projects were based on internationally recognized standards for offsetting such as the American Carbon Registry (ACR), Climate Action Reserve (CAR), Gold Standard (GS), and Verra's Verified Carbon Standard (VCR) (See *Appendix C* for a list of verifiers). In their small sample of international voluntary projects with co-benefits, they note that half focused on forestry/land-use projects, 30% focused on household device distribution, 10% energy efficiency/fuel switch projects, and 5% were engaged in renewable energy (Kountouris et al., 2014). Moreover, the projects in this study occurred in developing countries in Africa, Latin America, and Asia, which may differentiate them from projects in a developed country; however, some of the projects were in North America.

While these projects were not necessarily indicative of all of the project types that are taking place on the voluntary international offset market, they argue that the sample suggests these projects result in co-benefits. These co-benefits were employment creation, sourcing of local materials and services, household savings, capacity building, infrastructure development, ecosystem improvement, increased biodiversity, technology transfers, stakeholder participation, improved public health, improved air quality, and improved water sources (Kountouris et al.,

2014). Offsets from the international market have the potential to facilitate sustainable development, particularly by supporting communities in their adoption of low-carbon technology, supporting public health, alleviating poverty, and promoting ecological conservation (Kountouris et al., 2014). The latter “co-benefits” are the additional socio-economic, health, and environmental benefits that result from offset projects beyond that of greenhouse gas reduction (Kountouris et al., 2014).

The empirical research on the co-benefits of voluntary offsets is still emerging and there is limited existing research on their specific valuation. Moreover, the quantification of co-benefits can be risky, especially depending on whether the quantification is conducted ex-ante or ex-post, particularly due to unforeseen difficulties in implementation of a project (Kountouris et al., 2014). The monitoring and quantification of co-benefits need expansion to help institutions make decisions about their carbon offset projects. Since quantification is not currently widely adopted this may mean that co-benefits to voluntary offsets are being undervalued. In addition to direct co-benefits, some research is suggesting that co-benefits can also have “ripple effects” (Kountouris et al., 2014). For example, voluntary offset projects may have secondary impacts such as increased job creation in the local supply chain (Kountouris et al., 2014).

In discussions with the PWG, it became clear that offsetting the university’s carbon emissions deficit via local interventions is a priority of the PWG leadership, and that equity and quality of life improvements for recipients of projects was central to the university’s carbon offset effort. Thus, considerations of co-benefits are central to carbon offset proposals that were developed throughout this research project.

Methodology and Criteria

The goal of this project is to examine possible alternatives for actions that USC may take in order to reach its goal of carbon neutrality. Researchable questions were developed at the beginning of the process, and these became the preliminary research questions for defining the extent and scope of the project, and for framing the underlying assumptions in conducting the analysis.

These questions were:

1. What are the criteria used to evaluate offsets and carbon-reduction projects? What types of projects are cost-effective ways to offset carbon locally? What projects are most effective in reducing carbon emissions?
2. What are urban areas and universities doing to offset carbon (i.e., types of projects or technologies being used)? What are their impacts?

3. Based on community feedback and expert perspectives, how can USC's offset plan best appeal to the neighborhoods surrounding campus? What opportunities for partnerships exist?

To be more specific, we support the analysis incorporated in this report with a literature review and interviews with university counterparts and other experts with whom we were able to interact, USC faculty including those on the PWG, and key stakeholders in USC neighborhoods. Primary research and stakeholder engagement served to support the preliminary assessment of community preferences and needs, and the feasibility of the alternatives and threats to alternatives' implementation and/or efficacy. The boundaries of USC's Good Neighbors Campaign were used to guide the jurisdiction within which options would be put into place. We were also able to host a listening session with members of these communities. The community listening session and other interviews were the primary sources of first-hand information from the community.

In addition to the above methods, the team utilized financial- and cost-benefit analysis to deepen the understanding of this project and, more importantly, to determine how to scale the policy options that were explored in a local context. This includes the use of Monte Carlo sensitivity analysis to account for uncertainty. While exploring alternatives, the team researched examples of carbon offset co-benefits in the literature, with special emphasis on urban and university projects with local or regional benefits. These co-benefits included examples of economic development, generation of employment opportunities, health improvements, and educational opportunities. This approach provided examples of how projects were brainstormed, created, and implemented with community input. The team also gathered information on trends in city-university partnerships. By examining the literature on effective practices as well as actual programs developed by cities and universities, the team narrowed down options based on models of non-carbon impacts. This helped ground an understanding of typical projects and their estimated benefits to support an *ex-ante* cost-benefit analysis that sums the expected carbon reduction resulting from alternative projects and subtracts their costs (Boardman et al., 2010). It is important to note that while we intended to quantify and monetize the co-benefits stemming from each of the options, this was not always the case due to significant limitations in the empirical literature. For this reason, we chose, in those instances, to report the associated co-benefits from each project alongside net values.

Throughout this report, carbon offset options are valued against the baseline counterfactual of carbon offsets purchased on an international offsets market. The assumed price for these offsets is \$15/metric ton CO₂, or a baseline cost of \$1.38 million to offset USC's Scope 1, Scope 2, and Scope 3 emissions (92,000 metric tons CO₂). The market price of \$15/metric ton CO₂ was supplied to the practicum team by the client, in a series of interviews occurring in the Fall of 2020.

Carbon emissions reduction benefits are monetized throughout this report to inform the net present value of carbon offset investments. The standard measure for carbon emissions valuation in cost-benefit analysis literature is the Social Cost of Carbon (Boardman et al., 2018). In some cases, this is an inherently political figure - the Biden administration, for example, recently set the federal Social Cost of Carbon to \$51/metric ton CO₂, up from the Trump administration's \$1/metric ton CO₂ (Chemnick, 2021). In this report, we use a Social Cost of Carbon figure set at \$54.50/metric ton CO₂. This is the average of the proposed Social Cost of Carbon in 2025 and 2030 – \$52/metric ton CO₂ and \$57/metric ton CO₂, respectively – from the California Air Resources Board's 2017 Climate Change Scoping Plan (CARB, 2017).

Criteria-Based Analyses of Carbon Offset Options

Carbon offset options considered in the development of the University's carbon offset strategy are subject to a standard criteria-alternatives analysis. Analysis under a criteria-alternatives foundation allows each carbon offset option to be considered in direct comparison to one another. Options are listed in a matrix along with certain criteria, and the carbon offset options are rated according to a predetermined metric of fit for each criterion. Carbon offset options can be assessed under this quantitative method for the "best" option at face value. While a criteria-alternatives matrix ("CAM") provides insight into carbon offset options that best satisfy criteria for inclusion in the university's carbon offset strategy, more detailed analysis is required into the full costs and benefits of each option.

The criteria-alternatives analysis performed in the development of the final recommendations for inclusion in the University's carbon offset strategy evaluates carbon offset options under five standard criteria: Cost, Feasibility/Efficiency, Co-Benefits to Community, Emissions Reduction Potential, and Equity. These criteria were selected by the practicum team following a series of initial interviews in the Fall of 2020 with the USC Presidential Working Group on Sustainability that illustrated the client's values and goals for the university's carbon offset strategy. Based on these interviews, the practicum team determined that any recommendation for the University's carbon offset strategy must meet, at the very minimum, a total criteria score of "12."

A secondary criteria-alternatives matrix is included that weighs potential carbon offset options against the industry-standard P.A.V.E.R. framework utilized for compliance purposes by the California Air Resources Board. This framework assesses carbon offset options under the following criteria: Permanent, Additional, Verifiable, Enforceable, and Real (California Health and Safety Code §38562(d)(1) and (d)(2)). The five standard criteria selected for inclusion in the primary criteria-alternatives analysis, as well as the five criteria of the P.A.V.E.R. framework, are explained in detail below.

Explanation of Criteria: Primary Analysis

The following section discusses assignment of a subjective numerical scale for evaluating our criteria. Future research could develop a more rigorous approach for analysis of project options.

Cost: The Cost criterion measures carbon offset options in terms of absolute monetary cost. Costs are reported as the net present cost of each carbon offset option. For the purposes of the criteria-alternatives analysis, carbon offset options are assigned a rating based on net present cost between 1 and 5, with 1 being “very high cost” and 5 being “very low cost.” Given the lack of a predetermined budget for the University’s carbon offset strategy, ratings are assigned to carbon offset options in comparison to the cost of the baseline option, carbon offset purchases on an international market. As discussed in the *Methodology* section, the baseline cost is \$1.38 million. Ratings of carbon offset options under the Cost criterion are consequently assigned:

- 1 (“very high cost”) if the net present cost is greater than or equal to \$2.75 million
- 2 (“high cost”) if the net present cost is between \$2.25 million and \$2.75 million
- 3 (“moderate cost”) if the net present cost is between \$1.75 million and \$2.25 million
- 4 (“low cost”) if the net present cost is between \$1.38 million and \$1.75 million
- 5 (“very low cost”) if the net present cost is less than or equal to \$1.38 million

Feasibility/Efficiency: The Feasibility/Efficiency criterion measures carbon offset options based on ease of implementation. While this criterion factors in monetary cost, the emphasis of the Feasibility/Efficiency criterion is project implementation. Metrics considered in the assignment of ratings to carbon offset options under this criterion include the degree of projected community buy-in, time to complete project implementation, availability of project resources, bureaucratic processes for project implementation, and cost-effectiveness. Ratings are assigned to carbon offset options as a summary variable from 1 to 5, with 1 being “very weak” and 5 being “very high.” The process for determining summary ratings involves a degree of subjectivity. The Feasibility/Efficiency criterion assignments are as follows:

- 1 (“very weak”) if project implementation is determined to be especially difficult
- 2 (“weak”) if project implementation is determined to be somewhat difficult
- 3 (“moderate”) if project implementation is determined to be non-challenging
- 4 (“strong”) if project implementation is determined to be somewhat simple
- 5 (“very strong”) if project implementation is determined to be especially simple

Co-Benefits to Community: Initial interviews with the USC Presidential Working Group on Sustainability, conducted by the practicum team in August and September of 2020, outlined an important goal of the practicum’s analytic project: to determine if a carbon offset strategy could be enacted by the University that provides substantial co-benefits to the communities surrounding USC’s University Park Campus and Health Sciences Campus. Following this mandate, the practicum team has included Co-Benefits to Community as a criterion in the criteria-alternatives analysis. The Co-Benefits to Community criterion measures the number of

co-benefits associated with each carbon offset option. Ratings are assigned under this criterion from 1 to 5, with 1 being “very low” and 5 being “very high.” The complete Co-Benefits to Community criterion ratings are as follows:

- 1 (“very low”) if there are zero co-benefits to the community
- 2 (“low”) if there is one co-benefit to the community
- 3 (“moderate”) if there are two or three co-benefits to the community
- 4 (“high”) if there are four co-benefits to the community
- 5 (“very high”) if there are five or more co-benefits to the community

Emissions Reduction Potential: Carbon emissions deficit reduction via carbon offsets is the primary goal of the practicum’s recommendations for a university carbon offset strategy. Therefore, Emissions Reduction Potential is an important criterion for inclusion in the criteria-alternatives analysis. Carbon offset options are rated based on a comparison to the projected USC annual carbon deficit of 92,000 metric tons. However, since the reduction of 92,000 metric tons of carbon emissions annually is expected to require a strategy that includes more than one carbon offset option, the “moderate” level of carbon emissions reduction from one carbon offset option is set at a lower median target of 30,000 metric tons. Ratings are assigned under the Emissions Reduction Potential criterion from 1 to 5, with 1 being “very weak” and 5 being “very strong.” The complete Emissions Reduction Potential criterion ratings are as follows:

- 1 (“very weak”) if the annual carbon emissions reduction potential is less than or equal to 15,000 metric tons
- 2 (“weak”) if the annual carbon emissions reduction potential is between 15,000 metric tons and 25,000 metric tons
- 3 (“moderate”) if the annual carbon emissions reduction potential is between 25,000 metric tons and 35,000 metric tons
- 4 (“strong”) if the annual carbon emissions reduction potential is between 35,000 metric tons and 45,000 metric tons
- 5 (“very strong”) if the annual carbon emissions reduction potential is greater than or equal to 45,000 metric tons

Equity: The final criterion for inclusion in the primary criteria-alternatives analysis is Equity. While the Co-Benefits to Community criterion measures absolute benefit to USC’s neighboring communities from the University’s carbon offset strategy, the Equity criterion measures the distribution of benefits. Ratings under the Equity criterion measure the extent to which all members of the community reap the co-benefits associated with USC’s carbon offset strategy, regardless of age, race, access, or other demographic factors. Ratings are assigned to carbon offset options as a summary variable from 1 to 5, with 1 being “very weak” and 5 being “very high.” The process for determining summary ratings involves a degree of subjectivity. The complete Equity criterion assignments are as follows:

- 1 (“very weak”) if the distribution of co-benefits is determined to be especially narrow
- 2 (“weak”) if the distribution of co-benefits is determined to be somewhat narrow
- 3 (“moderate”) if the distribution of co-benefits is determined to be generally acceptable
- 4 (“strong”) if the distribution of co-benefits is determined to be somewhat broad
- 5 (“very strong”) if the distribution of co-benefits is determined to be especially broad

Explanation of Criteria: P.A.V.E.R. Framework

The P.A.V.E.R. framework for compliance of carbon offsets is advanced under California Health and Safety Code §38562(d)(1) and (d)(2). The P.A.V.E.R. standard criteria are: Permanent, Additional, Verifiable, Enforceable, and Real. The Permanent criterion tracks the long-term effectiveness of carbon offset projects and ensures the California Air Resources Board does not credit organizations with carbon offsets for intermediate carbon emissions reductions. Carbon offset options satisfying the Permanent criterion are determined to feature zero risks of future emissions over the project time horizon (GAO, 2008). The Additional criterion measures the actual impact of carbon offset projects. Carbon offset options that satisfy the Additional criterion certify that carbon emissions reductions would not have occurred in absence of the carbon offset intervention (GAO, 2008). The Verifiable criterion certifies that carbon offset claims by an organization are subject to review by an independent authority (GAO, 2008). The Enforceable criterion tracks “exclusive ownership,” protecting against double-counting of carbon offsets – one metric ton of carbon emissions reduction must be counted as one metric ton of carbon emissions reduction, regardless of position in the supply chain or scope of carbon emissions (GAO, 2008). Finally, the Real criterion ensures carbon offsets reflect a real net reduction of carbon emissions (GAO, 2008). Given the binary nature of criteria in the P.A.V.E.R. framework, the secondary criteria-alternatives analysis rates carbon offset options with 0 if the project does not satisfy P.A.V.E.R. criteria and 1 if the project does satisfy P.A.V.E.R. criteria.

Strengths & Weaknesses of Project Methodology

The following section describes the strengths and weaknesses of the project’s methods and the considerations for USC’s Presidential Working Group as it explores carbon offset strategies.

Strengths of Policy Options Analysis and Feasibility Analysis

To develop the offset options, the team placed a significant emphasis on individuals, households, and small commercial businesses as focal points of intervention. This focus was primarily due to conversations with the members of the Presidential Working Group and their interest in ensuring that residents near USC’s campuses would reap benefits from investments the University made to offset its carbon emissions. Indeed, even in the carbon offset protocols provided by influential carbon accounting and offset standards agencies, such as VCS or ACR, it is not apparent that there are many examples of individuals or households in North America as the loci of

intervention or as beneficiaries of market-based projects. Thus the present study has had a crucial philanthropic lens on the options to be considered, with a significant focus on innovation.

In considering options' feasibility, the present project also focused significantly on faculty, staff, and community expert opinions, via semi-structured interviews. The project team was able to meet on a biweekly basis with the PWG and schedule numerous semi-structured interviews for input on viable options and examine the existing partnerships between individuals and departments at USC as well as external organizations or initiatives. Because of the close work with experts, the project team could leverage information on emerging USC campaigns, including the current focus on developing solutions to food access and insecurity during the COVID-19 pandemic.

Because localized voluntary carbon offsetting is a relatively new concept, as most offsets take place internationally, where the cost per metric ton reduced is significantly cheaper, the present study reviewed verified carbon offset methodology and protocols without necessarily constraining the possibility of options to what was already in existence. Despite the limited examples of localized carbon offsets from which to draw inspiration, it became apparent very quickly that our interviewees from within and outside of the University were excited about the prospect of such initiatives. This research validates that there is significant potential for USC's faculty, leadership, and staff to develop new research, methodologies, and carbon offset protocols while benefiting the residents in the neighboring communities.

Focusing on the internal USC community's insight into potential projects meant that the project could uncover multiple offices, academic faculty members, departments, and partnerships that the University could form to make localized offsetting a reality. It became clear that the researchers on campus are already concentrating on issues related to these topics and that community relations offices within the University are also working on supporting households and families in surrounding communities. With funding, support, and leadership investment, emerging offset initiatives could create significant learning and research opportunities for students, faculty, and the local community.

Vulnerabilities of Policy Options Analysis and Feasibility Analysis

The applied portion of this project took place over one academic semester, during the COVID-19 pandemic. There were many policy options that were not explored in depth due to time constraints and extenuating circumstances from the COVID-19 pandemic, but which we believe merit further attention. These offsets included more community-based – rather than household-based – interventions, including offsetting through urban forestry. Other offset project types that are more technologically based or ecological were also overlooked. Because these types of offsets, such as methane destruction and sequestration/utilization projects or ecological

projects that feature restoration of the natural environment (i.e., wetlands or forests) tend to have more backing in terms of examples in the international protocol literature, USC's PWG may want to explore these types of projects further.

It will be necessary as an additional building block from this work for USC to determine whether the institution prefers third-party verification (which would allow it to eventually put offset carbon credits on the market) or peer review of the initiatives it seeks to pursue (which will lend to credibility and partnership with peer institutions or agencies). Moreover, it appears that, recently, universities' strategies for developing offset projects have been pushing for determining offset projects through requests for proposals (RFP) processes. Faculty experts and student supporters submit suggestions or ideas for local pilots that offset the carbon that they may also be interested in developing, field-testing, or evaluating. For example, in the summer of 2019, the University of California requested RFPs from students and faculty, with a particular interest in supporting local communities near their campuses, and notably framing offset investments as investments in both impact and research. Similarly, according to Duke Carbon Offsets Initiative's 2020 annual report, Duke has also begun developing an evaluation tool and RFP process to help solicit and compare offsets from the carbon market to ensure future purchases align with its priorities and standards. It may be beneficial for USC to consider generating ideas in a similar fashion, which falls outside the parameters of this report.

USC must similarly develop priorities around its approach to higher-risk versus lower-risk offsets, given the variability in quality associated with different offset options. For example, some sources imply that offsets with high co-benefits tend to be riskier than those with lower co-benefits (The Carbon Offset Research and Education Program, 2021). Similarly, some offsets have a more challenging time meeting the essential carbon offsets criteria and demonstrating environmental integrity than other options (i.e., P.A.V.E.R.), which beckons the University to create strategic priorities for the environmental integrity of projects to pursue.

Regarding the method of multicriteria decision analysis used in this study, it must be noted that aggregated criteria ratings yield final recommendations that may be highly subjective and also disregard certain stakeholder groups under unfavorable conditions (Kiker et al., 2005). One such unfavorable condition is an arbitrarily high minimum rating for inclusion in the final set of recommendations. While it may seem prudent to select policy options based on a strict minimum, such as a score of "15" (a minimum of "3 out of 5" rating for each criterion), the floor of "12" used in this study allows incommensurable criteria to be weighed against one another based on value to various stakeholder groups. Setting the floor at "12" allows for variation in ratings among the five criteria, leading to a set of recommendations that is not so poorly affected by the negative aspects of multicriteria decision analysis.

In terms of communications of costs surrounding the projects under consideration, we found high variability from both advocacy papers and empirical studies. This means that our estimates must

be taken with caution. Additionally, the costs derived for each project must be made more comparable in terms of financing strategies to avoid misrepresentation. We also recommend that field testing and further approximation take place to make judgment calls on the cost-effectiveness of projects to pursue. Moreover, because of the lack of a budget constraint for this project, we may have considered a range of possibilities that are not actually preferable or even feasible if and when a budget for USC's localized offsets emerges in the future.

Lastly, we felt our community engagement was limited to taking place through the gracious support of the USC Office of Community Relations. This on its own is not a weakness, as we were able to hold one listening session with community stakeholders. However, we believe that future work could leverage community partnerships and authentic community engagement through a community needs assessment that determines what community leaders, partners, and members in South and East Los Angeles would like to see in their communities. Future work must engage with community stakeholders and partners to a higher degree than we were able to achieve.

Analysis and Findings

The following section describes the findings and preliminary estimates regarding different carbon offset options.

Carbon Offset Options: Residential Strategies

By concentrating on developing residential energy efficiency projects for people within the boundaries of the Good Neighbors Campaign, USC has the opportunity to use a host of well-researched residential interventions that have positive outcomes, including reducing significant energy burdens on vulnerable populations. Residential energy efficiency projects have received significant academic attention due to their potential for delivering health and pollution mitigation benefits. In this report, residential strategies include two options 1) weatherization of single-family homes and 2) solarization of single family homes.

Residential weatherization typically refers to interventions that reduce energy use due to 1) building envelope improvements, 2) appliance upgrades, and 3) all-energy systems interventions. For this report, residential weatherization will refer to a project sponsored by the University that provides low-income households in the community with energy audits and supports their follow-through on the subsequently recommended interventions by covering the cost of the improvements up to about \$5000. An interview with and documentation from Duke University's Carbon Offsets Initiative (DCOI) demonstrated that weatherization of residences could be a form of offset project, particularly since a methodology exists for quantifying the GHG emissions reductions resulting from implementing measures that reduce the energy consumption of both single- and multi- family buildings (VM0008 Weatherization of Single Family and Multi-Family Buildings, v1.1). DCOI's Home Energy Loan (HEAL) program offered free home energy

assessments, which were covered financially by the university, to a small group of university employees, who subsequently had access to low-interest loans for implementing the recommended interventions, which resulted in 15-20% reductions in energy usage on average per household after implementation. Since this report's hypothetical project would target low-income homes, it was assumed that USC would recruit single-family households from South and East Los Angeles, connect them with an energy auditor, cover the cost of the audit, and pay for the cost associated with recommended energy efficiency improvements at a maximum of \$5,150 dollars per household.

Several studies suggest that residential solar has inequitable take-up because of high upfront costs, complexity of purchasing a system, and time required to install. For this project, a hypothetical offset project would target and recruit single-family households in South and East Los Angeles with average annual consumption in energy for installation of photovoltaic (PV) systems. These residences would have significant barriers (both financial and logistical) to transitioning to solar installation (perhaps they are low-income) without USC support, must have the appropriate roof qualities to make installation feasible, and plan to own/live in their homes for at least 25 years into the future. In this hypothetical project, USC would pay for the cost of the solar panels and their installation and claim offset credits resulting from the emissions reductions for the lifetime of the installation. This hypothetical project would focus on home-owners since it would mean avoiding going through a landlord or intermediary and could also build a partnership with a solar contractor to negotiate a group discount.

Additionally, the participants of either of these programs would need to provide utility use information for at least a year prior to implementation, for the purposes of tracking pre- and post-installation energy savings, developing the carbon reductions estimates, and for verification purposes. Participants must agree to the upkeep or maintenance of the installations and to not undo any of the improvements made.

Residential Weatherization

The following section outlines residential weatherization findings based on the criteria of cost, feasibility/efficiency, reduction potential, co-benefits, and social equity.

Cost of Weatherization Projects

DCOI's HEAL program ran from 2012-2016, and was implemented among Duke's employees; it cost approximately \$1,000 per employee to implement the Duke COI HEAL program. Based on the HEAL program's findings, DCOI estimated that, on average, the cost of avoided carbon was \$124 per metric ton and that the project avoided about 1.5 mtCO₂e per house per year over the span of 10 years.

The first large-scale field study on the returns to energy efficiency investments by Fowlie, Greenstone and Wolfram's (2018) on the Weatherization Assistance Program (WAP) suggests that the direct costs of weatherization projects may significantly exceed energy benefits. Their study, which consisted of a randomized encouragement design, examined a sample of approximately 30,000 low-income households in Michigan from 2011 to 2014. Each household, on average, received \$5,150 worth of home improvements, paid nothing out-of-pocket, and most commonly underwent furnace replacements, insulation in walls or attics, and reductions in the infiltration of their homes (Fowlie, Greenstone, and Wolfram, 2018).

The authors argued that the existing literature and engineering models for WAP might overestimate some energy efficiency investments' cost-effectiveness, demonstrating the importance of field testing such investments (Fowlie, Greenstone, and Wolfram, 2018). Because the WAP Program only pays for improvements that pass an ex-ante cost-benefit test (i.e., where benefits outweigh costs based on engineering projections), the study's findings were surprising to many in the residential energy efficiency space. The energy efficiency improvements were estimated to have reduced energy consumption by 10-20% in recipients' homes (Fowlie, Greenstone, and Wolfram, 2018). Thus, the authors assert that a reasonable average cost per ton of avoided carbon for the WAP program implemented was \$200 per metric ton.

In this analysis, the average annual consumption of a household near USC's campuses was assumed to be 6,684 kWh, based on the California Public Utilities Commission (2015) metric that, on average, the typical California home uses around 557 kWh per month (See *Table 1*). The cost of improvements was set at \$5150, taking from the average estimated in the Fowlie, Greenstone, and Wolfram study (2018) (See *Table 1*). While in reality, the cost of an energy audit depends on many factors, including the square footage of a residence, for the purpose of this report, the cost of an energy audit was set at the average, which is approximately \$409 (Home Advisor, n.d.) (See *Table 1*).

Table 1

Average annual consumption in kWh (CPUC, 2015)	6,684
Hours in one year	8,760
Average annual consumption in kW	3.5
Average cost of home improvements per household (Fowlie et al., 2018)	\$5,150
Average cost of residential energy audit (Home Advisor, n.d.)	\$409
Outreach/Recruitment Costs per Household (Fowlie et al., 2018)	\$1000
Average energy savings per household (DOE, 2021)	15%
Projected lifespan of energy efficiency improvements (Fowlie et al., 2018)	10 years

Energy Benefit & Co-Benefits

According to the study by Fowlie et al. (2018), weatherized households reduced their energy consumption by about 20% (Fowlie, Greenstone, and Wolfram, 2018). The resulting savings were \$2,400 over the lifespan of the energy efficiency measures installed (Fowlie, Greenstone, and Wolfram, 2018). However, while the energy savings analyzed were small in this particular study, other researchers suggest that the energy savings can be up to \$9000 over ten years, and that there are likely many non-energy savings benefits to weatherization.

For this analysis, the energy savings resulting from weatherization upgrades were set to 15% per home per year, based on the Department of Energy's findings that the annual energy savings from a WAP home ranged from 10% to 20% per home on average per year (DOE, 2021) (See *Table 2*). Whereas before weatherization, a hypothetical typical home would emit approximately 1.52 mtCO₂e per year, after weatherization, with 15% reduction in energy savings, we thus estimated a 0.23 mtCO₂e reduction in emissions per household per year (See *Table 2* for a depicted of before and after emissions from energy use of a single household). Using the Social Cost of Carbon, which this project set at \$54.50 per ton of CO₂, the weatherization of one household would provide social benefits (from emissions reductions alone) worth \$12.53 per year. Moreover, the Verified Carbon Standard (VCS) protocol for offsetting via weatherization uses a 10-year life span for analyzing these types of all-energy efficiency retrofits. After 10 years of the projected lifespan of the installed improvements, one weatherized home would reduce an approximate total of 2.3 mtCO₂e. Moreover, using the CPUC's cost estimate of \$0.16 cents per kWh, an energy consumption reduction of 15% resulting from weatherization could save a household approximately \$160 per year on their energy bill. For the purpose of this project, we

did not include the bill savings from weatherization to develop a CBA, but future investigations should include these benefits for increased depth of analysis.

Table 2

Energy Source	Proportion of CA energy (%)	Tons CO₂/k Wh (generation only)	Carbon emissions of one household (before weatherization)	Carbon emissions of one household (after weatherization)
Natural gas	54.6	0.00041 2769	1.506385606	1.280427765
Solar	21.3	0	0	0
Wind	8.1	0	0	0
Hydroelectric	7	0	0	0
Nuclear	5.5	0	0	0
Coal	0.2	0.00100 2439	0.01340060455	0.01139051387
Total mtCO₂e per house per year:			1.51978621	1.291818279

Feasibility/Efficiency of Weatherization Projects

Several authors suggest that one significant problem with weatherization projects is the difficulty in getting low-income households to participate, even when there is no out-of-pocket cost to them, such as through the WAP program. Fowlie, Greenstone, and Wolfram (2018) mention that their study had low sign-up rates despite the significant marketing efforts and dollars spent in encouragement to participate. Field activities for the recruitment of households involved an intensive outreach campaign, involving home visits, phone calls, and multiple follow up-appointments, at an approximate cost of \$1000 per weatherized household (Fowlie, Greenstone, and Wolfram, 2018) (See estimated cost of recruitment/outreach in *Table 2* above). The authors state that the small differences in uptake between the sample encouraged to participate and the sample that was not, reveal a low demand for weatherization programs, despite the purported benefits. However, differences in cost of living, utility rates, and space heating/cooling needs vary broadly by region and other factors such as differences in climate, characteristics of housing stock, and energy production mix highly impact the cost-effectiveness of specific weatherization treatments (Bradshaw et al., 2016). For example, Bradshaw et al. (2016) posit that people living in low-income homes in milder climates such as in Long Beach or

Los Angeles may be more interested in programmable thermostats, which are more cost-effective and relevant than other treatments like attic insulation or air sealing.

It is vital to note that the efficiency of weatherization treatments for energy savings are under considerable debate. For example, some engineering models used to justify the WAP Program, for example, have predicted energy savings of more than \$9000 per household over the projected lifespan of energy efficiency improvements installed, which tend to be an average of 10 years (Fowlie, Greenstone, and Wolfram, 2018). Thus, it is important to acknowledge that there are different estimates in the literature for residential weatherization treatments.

Co-Benefits from Weatherization Projects

It is important to note that this report did not include the monetized non-energy savings benefits which may be significant and especially beneficial for the most disadvantaged populations. These included other impacts from weatherization, such as higher home values and increases in localized job creation (Bradshaw et al., 2016). Some sources suggest weatherization can also improve indoor air quality, reduce respiratory issues, increase the comfort of a home (i.e., by enhancing them from drafts and leakage), and reduce the emissions from local power plants due to decreased demand (Adair, Elliot, and Weiss, 2016).

Reduction Potential for Weatherization Projects

In the approved Verified Carbon Standard (VCS) offset protocol literature, the process for estimating weatherization projects' emissions reductions vary depending on the type of project. For "all-energy" retrofit projects and residential building enhancement projects (such as WAP), calculating the emission reductions and monitoring such reductions tend to be based on either the adjusted consumption approach, the pre- and post-retrofit audit approach, or the control group approach (VCS, 2011). For this project, we assumed that this program would use a pre- and post-weatherization approach.

In order to offset 10,000 mtCO₂e per year for 10 years, USC would need to weatherize 4,348 single-family homes at an upfront cost of \$6,559 per house, or net present cost (NPC) of \$28,062,889. In terms of carbon reduction benefits and bill savings benefits for the households in question, this would generate a present value of benefits of approximately \$1,446 each household, or \$6,286,710 (PVB, including reduction in emissions and reductions in energy bills) across all targeted households. Assuming 10,000 metric tons of avoided carbon per year for ten years, the cost per avoided carbon would be \$280 per metric ton.

However, this analysis did not account for monetized co-benefits that would result such as those from improved indoor air quality, increased home value, increased comfort, decreased local pollution from energy combustion, or localized green job creation. It also assumed that USC

would target only single family homes, pay for all of the total cost of weatherizing a home upfront (including the marketing/outreach costs), and that each household would only reduce its energy consumption by 15% per year for the lifespan of the installations. Our conversations with a representative from the ACEEE suggested that educational and behavioral interventions may further impact the energy savings resulting from these types of energy efficiency improvements, which means the assumption of 15% reduction in energy usage per year and household must be taken with caution.

Table 3

Residences weatherized to reach 10,000 mtCO ₂ e per year	4,348
Annual carbon emissions reduction potential (metric tons CO ₂)	10,000
Cost (present value)	\$28,518,532
Carbon emissions reduction benefit (present value)	\$455,643
Net present cost	\$28,062,889
Cost effectiveness per metric ton reduced	\$280

Equity for Weatherization Projects

In terms of estimated need in the communities surrounding USC, there are likely numerous households within the Good Neighbors Campaign boundaries that need significant improvements to energy efficiency via weatherization. Many documents describe that low-income people face several barriers to making energy efficiency upgrades to their homes. The renter versus owner disparity is one factor, but lack of capital, knowledge, or information may also drive these disparities, even when public-funded programs are available. According to data from the US 2018 US Census Bureau, in the zip-codes surrounding the HSC campus, approximately 34% of the housing was built in 1939 or earlier. Meanwhile, in zip-codes surrounding UPC, about 44% of housing was built in 1939 or earlier. Thus, it is plausible that people living in significantly older homes in the communities surrounding USC could benefit from weatherization improvements.

Residential Solar

In our interviews with residential energy experts, it became clear that residential solarization had clear equity implications, particularly for this region in California. The following section outlines residential solarization findings based on the criteria of cost, feasibility/efficiency, reduction potential, co-benefits, and social equity.

Cost of Residential Solar Projects

The cost of a residential solar system depends on watts' size and the price per watt offered by installers (Behr et al., 2015). System size also tends to vary and is based on energy usage needs, space available on roofs, and adopters' financial limitations (Behr et al., 2015). The average household in California uses approximately 557 kWh monthly (Rockzsfforde & Zafar, 2015). This analysis assumes that the average CA household's annual consumption is 0.064 kW. Since the smallest rooftop solar panel typically starts at 2.5kW in size capacity, we used Mauritzen's estimate (2017) for median cost of installation per kWh, to assume the average PV installation cost for a household is \$14,325 (See *Table 4* below).

Table 4

Average annual consumption in kWh (CPUC, 2015)	6,684
Hours in one year	8,760
Average annual consumption in kW	2.5*
Median installation cost per kW (Mauritzen, 2017)	\$5,730
Median installation cost	\$14,325

*Note: The average CA household's annual consumption in kW is likely closer to 0.064 kW, but solar panels typically range from 2.5kW and larger in capacity.

Table 5

Energy Source	The proportion of CA energy (%)	Tons CO2/kWh (generation only)	Carbon emissions of one household
Natural gas	54.6	0.000412769	1.506385606
Solar	21.3	0	0
Wind	8.1	0	0
Hydroelectric	7.0	0	0
Nuclear	5.5	0	0
Coal	0.2	0.001002439	0.01340060455

Total Emissions from One Household:	1.51978621
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Feasibility/Efficiency of Residential Solar Projects

Several factors determine whether installing residential solar is suitable for households, one aspect of which is whether inhabitants are homeowners (Behr et al., 2015). This factor matters because a PV system tends to be a long-term investment fixed onto a home for about 25 years, and therefore may not always be considered worthwhile by renters or other people who plan to move after a short time (Behr et al., 2015). However, there are examples of organizations or communities that have negotiated with solar installers to drive prices down for interested multi-unit households. The organization GRID Alternatives in Los Angeles demonstrates that solar in multi-unit housing mostly occupied by low-income renters is possible, such as when the housing is owned or operated by nonprofit housing developers.

Furthermore, physical factors also determine whether a home is apt for solarization, including roof type, tilt, orientation, and shading (Behr et al., 2015). It will be vital for USC to understand the physical factors of individual homes and buildings near its UPC and HSC campuses to determine this intervention's feasibility for local people. Lukanov and Krieger (2019) point out that homes or buildings inhabited by low-income people may not always have the roofing space necessary to take advantage of solar installations, further aggravating solar adoption disparities. Furthermore, while solar community projects may be an attractive way to increase solar access in urban communities, they may be most viable when a significant industrial or commercial rooftop space is closeby and allows for large solar project installations (Lukanov and Krieger, 2019).

Another question that arises in the literature is whether lower costs due to solar installation may lead to a behavioral paradox known as a “compensation problem” (i.e., individuals/households using more energy when it costs them less). These concerns may warrant regular educational and behavioral interventions that are supportive of solar adoption's effectiveness and impacts.

Co-Benefits to the Community for Residential Solar Projects

Given the potential reductions in utility bills, whether individual or community, solarization may deliver financial benefits to households, especially when in poverty. For zip codes surrounding UPC, 28% of the population is in poverty, while this is 19% in HSC's surrounding zip codes (US Census Bureau, 2018). Our stakeholder interviews with community members and external relations staff highlighted recurring concerns with gentrification in East and South Los Angeles, potentially propelled by the presence and activities of the university. Because of these concerns, solarization interventions may be able to provide financial benefits to low-income renters and homeowners to help “keep” them in their homes. In 2020, the zip codes surrounding HSC were

estimated to be composed of 42% owner-occupied housing, while UPC's surrounding zip codes were estimated to be composed of 27% owner-occupied housing.

Research by Lukanov and Krieger (2019) state that solar adoption can lower electricity bills and provide co-benefits such as grid resilience (i.e., being able to have energy in the face of natural or human-made disasters) and reduce the local grid's emissions. In their study that analyzed cumulative and annual PV adoption across California, the researchers found persistently lower PV adoption levels in disadvantaged communities, indicating that the benefits of solarization may not be reaching some of the state's most vulnerable populations (Lukanov and Krieger, 2019). Since residential PV adoption can help lower electricity bills and make energy expenses more stable monthly, they argue that solarization may benefit households' ability to deal with other stressors beyond energy, including improving the ability to spend on their other health, education, and socioeconomic needs (Lukanov and Krieger, 2019)

Reduction Potential of Residential Solar Projects

In 2018, only .06% of occupied housing units in the zip codes surrounding UPC and .22% of such in the zip codes surrounding HSC ran on solar energy (US Census Bureau, 2018). Solarizing one household that consumes average energy per year would reduce 1.52 mtCO₂e per year, creating a lifetime savings of 38 mtCO₂e over the span of 25 years (See Table 5).

Table 5

Energy Source	The proportion of CA energy (%)	Tons CO₂/kWh (generation only)	Carbon emissions of one household
Natural gas	54.6	0.000412769	1.506385606
Solar	21.3	0	0
Wind	8.1	0	0
Hydroelectric	7.0	0	0
Nuclear	5.5	0	0
Coal	0.2	0.001002439	0.01340060455
Total Emissions from One Household:			1.51978621

Using the SCC of \$52.50, this annual reduction in emissions would generate \$82.82 in social benefits per year from emissions reductions and allow residents to save \$1069.44 per in energy bill savings (based on a cost rate of \$.16 per kwh from the CPUC, and assuming households do

not increase their energy use over the system’s capacity over time). Moreover, if they use less than the solar system generates, some families could even participate in LA’s Feed-in-Tariff Program and receive further financial incentives.

The present value of cost for solarization is estimated to be \$14,395 for one household, or \$94,709,670 to solarize approximately 6,579 homes to offset 10,000 mtCO₂e per year for 25 years (See *Table 6* below).. However, when accounting for the present value of the carbon emissions reduction benefit resulting, this would result in a net present cost total of \$87,198,587 (NPC) and the cost per metric ton of CO₂ avoided would be around \$348 (See *Table 6* below).

Table 6

Single-family residences transitioned to solar	6,579
Annual carbon emissions reduction potential (metric tons CO ₂)	10,000
Cost (present value)	\$94,709,670
Carbon emissions reduction benefit (present value)	\$7,511,082
Net present cost	\$87,198,587
Cost effectiveness per metric ton reduced	\$348

Equity of Residential Solar Projects

Lukanov and Krieger’s (2019) analysis shows that PV adoption is highly correlated with socioeconomic health and environmental indicators. They find a strong correlation of solar adoption in California communities by socioeconomic variables (i.e., education, language, race/ethnicity) but also with health indicators (i.e., asthma rates) (Lukanov and Krieger, 2019). These findings from the research on solar energy equity access suggests that this type of offset program may support the University’s equity values.

Carbon Offset Option: Small Business Solar Transition

Interviews conducted with Detlof von Winterfeldt and Stephanie Thrift of the USC Center for Sustainability Solutions (CSS) highlighted small business solar energy investments as another potential option for USC’s carbon offset strategy. According to CSS research, the total estimated carbon footprint of small businesses within the Good Neighbors Campaign boundaries of both the University Park Campus (UPC) and the Health Sciences Campus (HSC) of USC is about

244,000 tons CO₂ per year. This far exceeds the estimated 92,000 tons CO₂ deficit per year that USC seeks to offset through the recommendations of this practicum project. The carbon footprint of the businesses in each of the zip codes included in the Good Neighbors Campaign footprint is even larger when considering the full zip code – 3,740 small businesses host a total estimated carbon footprint of 748,000 tons CO₂ per year (von Winterfeldt and Thrift). Small businesses are defined by the research group as locally owned and operated franchises and businesses, not including large chain grocery stores.

Justification for the value of small business solar energy transition is provided by investigating the individual-level details of local small businesses' carbon footprints. The average small business in the Good Neighbors Campaign boundaries accounts for about 200 tons of CO₂ output per year (von Winterfeldt and Thrift). An estimated 75 tons of CO₂ per year comes from small business energy consumption (see *Benefits* section below). Therefore, solar energy transition in just one small business could offset up to 75 tons of CO₂ per year. In a best-case scenario where all 3,740 small businesses in zip codes that contain Good Neighbors Campaign boundaries agree to the solar energy transition, USC could offset up to 280,500 tons CO₂ per year.

When referring to “small business solar energy transition,” or variations thereof, the study focuses on a carbon offset option that involves USC paying for installation and/or maintenance of solar panels on the rooftops of local small businesses. There are opportunities for modification of this baseline carbon offset option, but the majority of this analysis is written with the perspective of a business-by-business solar panel installation paid for via the USC carbon offset budget. Suggested modifications for further research include loan programs that see USC pay the downpayment for solar panels and business owners adopt monthly payments, or an alternative where USC builds a central neighborhood solar facility that businesses can use to source clean power.

Cost Effectiveness of Small Business Upgrades

Costs

Energy efficiency upgrades in small businesses can produce long-term savings in carbon emissions. Data from the California Air Resources Board (CARB) and analysis from *The Energy Journal* provide insight into the cost, benefits, and cost-effectiveness of small business energy efficiency programs. According to the CARB Small Business Energy Calculator, the average small business in California consumes 329,158 kWh per year (CARB). An empirical analysis of solar panel installation costs in California found the median cost for installation between 2007 and 2014 was \$5,730 per kW (Mauritzen, 2017). Given the average energy consumption in kilowatt-hours of California small businesses, the average small business consumes about 37.6 kW per year. The median base year installation cost for solar transition in the average small

business in California is approximately \$215,000. Figures used in the calculation of the median solar transition cost are presented in *Table 7* below.

Table 7

Average annual consumption in kWh (CARB)	329,158
Hours in one year	8,760
Average annual consumption in kW	37.6
Median installation cost per kW (Mauritzen, 2017)	\$5,730
Median installation cost	approx. \$215,000

Benefits

According to Choose Energy, Inc., a national energy provider based in South Carolina, California sources 54.6% of its energy from natural gas, 21.3% from solar, 8.1% from wind, 7.0% from hydroelectric, 5.5% from nuclear, and 0.2% from coal (ChooseEnergy.com). Per the United States Energy Information Administration, natural gas produces 0.000412769 metric tons CO₂ per kWh of electricity and coal produces 0.001002439 metric tons CO₂ per kWh of electricity (U.S. EIA). Wind, solar, nuclear, and hydroelectric energy sources produce no carbon emissions during the electricity generation process, but life cycle assessments show each source contributes some carbon dioxide to the atmosphere during manufacturing and transportation of the equipment, as well as reservoir water storage at hydroelectric plants (Solar Energy Industries Association; World Nuclear Association; Ocko and Hamburg, 2019; Yang et al., 2015). For this analysis, carbon emissions reduction benefits are considered during the electricity generation process only – renewable energy sources are assumed to produce zero carbon emissions. Given the distribution of energy generation in California, solar energy transition in one small business would offset approximately 75 tons CO₂ per year. Valuing the carbon emissions reduction potential of small business solar transition using the Social Cost of Carbon at \$54.50/ton CO₂, solar transition in one local small business provides social benefits worth \$4,087.50. Figures used in the calculation of the average carbon offset benefit from small business solar energy transition are presented in *Table 8* below.

Table 8

Energy Source	Proportion of CA energy (%)	Tons CO ₂ /kWh (generation only)	Carbon emissions of one business
Natural gas	54.6	0.000412769	74.18
Solar	21.3	0	0

Wind	8.1	0	0
Hydroelectric	7.0	0	0
Nuclear	5.5	0	0
Coal	0.2	0.001002439	0.66

Cost-Effectiveness

Small business solar transition requires a \$215,000 up-front investment per business, but solar panel installation would offset 75 tons CO₂ per business per year over the functional lifetime of the solar panels at no additional cost. Given an average solar panel life expectancy of 27 years, an initial \$215,000 investment to transition one business to solar energy would generate a lifetime savings of about 2,025 tons CO₂ (Sunrun). Considering the long-term carbon offset benefits of small business solar transition, the cost-effectiveness ratio of this carbon offset strategy is approximately \$106 per ton CO₂ offset.

Financing and Tax Incentives

Solar transition can be financed under standard commercial loan programs and supported by a variety of federal and state tax incentives (DSIRE). For example, the federal Business Energy Investment Tax Credit provides a 26% tax incentive, up to a maximum of \$1,500 per 0.5 kW installed, for solar installation. Under a 15-year loan at 5.5% interest rate, with 20% down, USC would be liable for a down payment of \$43,000 and a monthly payment of \$1,405.38.

Alternatively, USC could implement a program where the university covers the down payment and small business owners are responsible for monthly loan payments. The present discounted cost of solar installation at one local business, using a 6% discount rate, is about \$66,000. With loan financing, the cost-effectiveness ratio of solar transition in local small businesses is approximately \$33 per ton CO₂ offset. This cost-effectiveness ratio and cost figure will be used in all further analysis given the likelihood that the university will pursue loan financing if small business solar transition is selected for inclusion in a final carbon offset strategy. Sensitivity analysis may be justified by adjusting the loan term and interest rate. The figures presented in this section on financing and tax incentives are summarized in *Table 9* below.

Table 9

Loan term (years)	15
Interest rate (%)	5.5
Down payment (20%)	\$43,000

Monthly payments	\$1,405.38
Present value of installation cost with loan financing	approx. \$66,000
Cost-effectiveness with loan financing	approx. \$33/ton CO2

Co-Benefits of Small Business Solar Transition

Solar energy transition in local small businesses yields non-monetized co-benefits to the communities surrounding USC. A 2019 study from *Nature Communications* finds that renewable energy systems, including solar energy, provide co-benefits by reducing environmental impacts from air and water pollution, land occupation, water use, ionizing radiation and nuclear waste, and fossil and mineral resource depletion (Luderer et al., 2019). These co-benefits derive from the reduction in reliance on non-solar energy sources. Additionally, given traditional economic theory on pass-through of costs to retail prices (Hong and Li, 2017; Miller, Remer, and Sheu, 2013; Weyl and Fabinger, 2013), we anticipate co-benefits to the local community may come from lower prices at market due to reduced energy costs for business owners that benefit from solar transition. Reduced cost pass-through may also increase demand, resulting in higher revenues for business owners and greater circulation of money in the local economy (Miller, Remer, and Sheu, 2013).

Small Business Solar Transition Offset Recommendation

Small business solar transition is a streamlined method to achieve high volumes of carbon offsets for a long period of time. Small businesses in USC's surrounding neighborhoods could contribute up to 280,500 tons of carbon offsets per year, at an average estimated cost of \$33 per ton over the 27-year average solar panel lifetime. The associated co-benefits of local commercial energy efficiency justify an investment strategy that includes small business solar transition. A potential recommendation could be to transition 400 businesses to solar energy for 30,000 tons of carbon offsets per year at a present discounted cost of \$26.4 million. The monetary social benefit of this recommendation is \$1,635,000 per year, or a present discounted value of about \$22.9 million. Note, this is only the monetary social benefit that comes from carbon emissions reductions. Monetary social benefits from co-benefits are not calculated as part of this analysis, but have wide-ranging economic implications that increase the overall monetary social benefit from this recommendation. The figures proposed in this recommendation are presented in *Table 10* below.

Table 10

Businesses transitioned	400
Annual carbon emissions reduction potential (metric tons CO ₂)	30,000
Cost (present value)	\$26.4 million
Carbon emissions reduction benefit (present value)	\$22.9 million
Net present cost	\$3.5 million

Healthy Food Market Investment

Another option available to the University is to invest in grocery stores and small marketplaces to improve the availability and accessibility of healthy food options in our surrounding communities. This may entail remodeling, adding shelf-/storage space, training, and otherwise boosting the capacity for businesses to stock and promote the sale of low-cost healthy foods like fresh fruits and vegetables.

Costs of Healthy Food Markets

The costs of implementing a healthy food markets program can vary significantly depending on the scope of the program. For instance, the difference between making use of existing infrastructure and building new marketplaces results in a very different cost structure. However, there is also a trade-off because of the lack of grocery store infrastructure in some parts of the communities, according to our stakeholder outreach. However, with over 1,300 small convenience stores across Los Angeles, it may make the most sense to create partnerships with existing stores and invest in their capacities instead of building new, competing stores (Nittle, 2019). Small market- or bodega revitalization projects have occurred in Los Angeles and in other communities in the U.S., such as New York City. The costs of revitalization with an emphasis on fresh fruits and vegetables in these stores can vary widely. For instance, in New York City, as a part of the Commercial Corridor Challenge, bodega remodeling has cost as little as \$20,000 per storefront (West, 2018). However, the Los Angeles Food Policy Council already has a program called the Healthy Neighborhood Market Network that does a host of similar work in Los Angeles. This includes structural revitalization, training for marketing and stocking fruits and vegetables, and other related investments (Los Angeles Food Policy Council, n.d.). These remodels vary considerably in cost, with examples ranging between \$65,000 and \$100,000 (Nittle, 2019). The USC Good Neighbors Campaign includes parts or all of eight postal codes.

These postal codes, each taken in its entirety, have a total land area of about twenty-five square miles. If the University were to invest in one market per every square mile, that would most likely cost between \$1,625,000 and \$2,500,000, using the per-store estimates of \$65,000 as a minimum and \$100,000 as a maximum as in the above similar projects in Los Angeles.

Feasibility/Efficiency of Healthy Food Markets

There is a wealth of historical programs focusing on healthy-eating interventions in communities around the United States. The scopes, goals, motivations, and outcomes of these programs and policies vary greatly, though most of them, if not all of them, have in common an emphasis on the health benefits for consumers (Karpyn et al., 2020). Public health interventions of this nature occur across rural, urban, suburban areas; in schools, markets, bodegas, and more; and derive funding from a variety of sources. As such, while there is no common or one-size-fits-all program off of which to base a new project, there is a wealth of data and resources that can inform related concepts and which indicate a high level of feasibility. New York City, alone, has undertaken numerous programs with similar emphases to that which we are recommending here (Sacks, Yi, and Nonas, 2015). That is, interventions with focuses on equity and access to healthy foods in low-income urban communities exist and provide a solid basis on which to build.

More specifically, there have been many programs that encouraged healthy dietary habits and improved perceived access to healthy options. Perception of availability of healthy options is important as it has been shown to increase consumption of those foods (Caldwell et al., 2009). One example is the Green Carts in New York City, which is a program designed to encourage the creation of mobile food carts that sell fresh fruit and vegetables that have reported positive results (Fuchs et al., 2014). The Healthy Neighborhood Market Network, too, has reported increases in profits and produce sales (Los Angeles Food Policy Council, n.d.). Other programs include a low-cost fresh produce market in Georgia and a program in New York to lower the cost of maintaining and owning grocery stores in certain areas (Woodruff et al., 2016) (New York City Economic Development Corporation, n.d.). As such, we can be confident that a healthy markets program in the study area is feasible and will create favorable outcomes. However, the feasibility of evaluating and verifying the offsets related to dietary changes from this program is murky. This would likely require serious thought and planning on the part of both the University and the businesses involved, and still, the best way to account for and quantify reductions is presently unclear.

Co-Benefits to the Community of Healthy Food Markets

As the name indicates, the main co-benefits to carbon reductions of healthy food markets are in encouraging and enabling healthier dietary habits in the surrounding communities. In our outreach to experts and community members, access to healthy grocery stores consistently

ranked as one of the most desirable investments and partnerships for the University to pursue. Currently, only 12.1% of adults in Los Angeles County self-report consuming the recommended five or more servings of fruits and vegetables per day (Los Angeles County Department of Public Health, 2018). That number is lower for Latino and African American residents, at 9.7% and 10.4%, respectively, who make up a large share of the population surrounding the University. In fact, for all of Los Angeles County, the average daily servings of fruit is 1.0 for all adults, 1.1 for White adults, 1.1 for African-American/Black adults, 0.99 for Latino adults, and 0.9 for Asian adults. The average daily servings of vegetables is 1.2 for all adults, 1.3 for White adults, 1.1 for African-American/Black adults, 1.1 for Latino adults, and 1.3 for Asian adults (de la Haye et al., 2021). Clearly, residents of Los Angeles County may benefit from more fruit and vegetable consumption. Action to encourage healthier dietary habits in the community is not only a benefit for the health of community members, but there is room for such an intervention with so few people consuming the recommended servings. Furthermore, this is a strongly desirable program among community members, based on feedback we received in interviews. Therefore, a partnership with the University and the community would be mutually beneficial and signal to the surrounding community that USC is listening wishes and willing to take action.

Emissions Reductions Potential of Healthy Food Markets

The largest share of carbon emissions per household in the area surrounding USC is due to diet (Thrift and von Winterfeldt, 2020). Findings by Stephanie Thrift and Detlof von Winterfeldt of the Presidential Working Group on Sustainability (PWG) inform this conclusion, breaking up the categorical greenhouse gas (GHG) emissions on average per household across postal codes (2020). The categories include transportation, housing, food, goods, and services. For a typical USC neighborhood postal code, food makes up the largest share of a household's carbon footprint. This includes all aspects of the supply chain for food and the composition of household diet in the area surrounding USC. As a comparison, the area around the UCLA area sees the largest share of household emissions coming from transportation, with food falling to fourth, after transportation, services, and goods (Thrift and von Winterfeldt, 2020).

A 2018 study catalogued the emissions that each of the most common foods contribute to the atmosphere (Poore and Nemecek, 2018). Using data from this study alongside USDA serving size recommendations (Chang, 2019) (Kantor, 1998) and OECD meat consumption data (OECD/FAO, 2020), we have estimated the emissions reductions potential of such an intervention. We were able to tell that, nationally, Americans consume approximately 120 grams of meat per day per person, leading to 4.6 kg of CO₂ equivalent extra per day per person. We used these data to estimate the emissions reductions of this program. First, we matched the grams-per-serving data to the corresponding food items in the Poore and Nemecek study. This table is available in *Table 11*. Then, we took the average emissions per gram of fresh fruits and vegetables as well as the average emissions per gram for beef, lamb and mutton, pig meat, and

poultry meat (*Tables 12 and 13*). A serving of fresh fruits and vegetables, for example, on average is 84.9 grams, and emits 1.7595E-06 metric tons of CO₂ equivalents.

We then conducted a Monte Carlo analysis to estimate, on average, how many metric tons of CO₂ this program could reduce. This decision was a result of the uncertainty around behavioral change; that is, we used the sensitivity analysis to account for the range of costs, the range of population uptake, and the variability in changes in purchases. The costs and the population are uniform variables, moving between a minimum of \$1,625,000 and \$2,500,000 and 0 and 375,000 (the population of the associated postal codes), respectively. For the change in purchases, we could expect weekly consumption of fruits and vegetables to increase by a mean of 2.971 servings with a standard deviation of 0.1128 servings (Caldwell et al., 2009). The parameters for this sensitivity analysis are in *Table 14*. To then calculate the emissions reduction, we took the changes in daily servings and converted it to grams of fruits and vegetables and grams of meat based on our grams-per-serving estimates for each. Then, we multiplied the emissions per gram of meat by the changes in servings of meat in grams to find the amount of CO₂ reduced by less meat consumption. We took these steps using the information in *Table 13*. Next, we multiplied the emissions per gram of fruits and vegetables by the changes in servings of fruits and vegetables in grams to find the amount of CO₂ added by more fruits and vegetables consumption. Finally, we calculated the net emissions change by subtracting the new emissions from fruits and vegetables from the reduced emissions from meat. This hinges on the assumption that people would be replacing meat with fruits and vegetables. This was computed for 10,000 iterations, and resulted in a mean of 125,515.83 metric tons CO₂ reduced, with a standard deviation of 91,044.38 metric tons CO₂ reduced. Using this average, we have an average cost per metric ton CO₂ of \$16.45. Using this figure, to reduce 92,000 metric tons CO₂ would cost \$1,513,400. Using the social cost of carbon estimate of \$54.50 from the *Methodology* section, this would result in a net present cost of -\$3,500,600 (in other words, it would result in net benefits). While this figure is attractive, the standard deviation of 91,044.38 metric tons CO₂ is nearly the amount of emissions reductions that is our target. Therefore, while potentially a very cost-effective option, healthy markets have an extremely high degree of variability in their outcomes. The results of this analysis are in *Graph 1*.

Equity and Healthy Food Markets

The PWG has already identified the equity considerations of a project in USC's surrounding communities. The annual household income in the USC area is \$62,108, compared to \$110,175 for the UCLA area (Thrift and von Winterfeldt, 2020). In addition, researchers have shown that lower-income households face barriers, whether perceived or real, to including plant-based foods in their diets (Leiserowitz et al., 2020). This is illustrative of the phenomenon whereby disadvantaged areas are more reliant on animal agriculture for their diet. Plant-based diets are generally healthier for people, including promoting lower mortality and lower risk of certain

diseases (Kim et al., 2019). In addition, plant-based diets produce lower GHG emissions than the traditional American diet (Lacour et al., 2018). With community desire for healthier food options and inequities further highlighted by the COVID-19 pandemic, the University has an opportunity to improve equity and access through this project.

Table 11 (FU = 1 kg retail weight unless otherwise noted)

Product		Grams per Serving	Mean GHG Emissions (kg CO ₂ eq/FU, IPCC 2013 incl. CC feedbacks)	Mean Emissions per Serving
Onions & Leeks	Vegetables	105	0.5	0.052500
Tomatoes		90	2.1	0.188100
Average Vegetables		64.80952381	1.3	0.083928
Bananas	Fruits	118	0.9	0.101480
Apples		138	0.4	0.059340
Citrus Fruit		106	0.4	0.041340
Berries & Grapes		78.5	1.5	0.120105
Other Fruit		84.57142857	1.1	0.088800
Average Fruits		105.0142857	0.852	0.089472
	Animal			
Bovine Meat (beef herd)	Products	155.925	99.5	15.511419
Lamb & Mutton		155.925	39.7	6.193341
Pig Meat		155.925	12.3	1.919437
Poultry Meat		155.925	9.9	1.538980
Average Meats		155.925	40.3	6.290794

*Note: *Other Fruit* uses the serving weights of apricots, avocados, cantaloupe, cherries, cranberries, honeydew melon, kiwifruit, mangoes, peaches/nectarines, pears, pineapple, papaya, plums/prunes, and watermelon and *Onions & Leeks* uses onions from Kantor, 1998.

Table 12

Emissions in kg CO ₂ eq per gram meat: Average Mean GHG Emissions per kg / 1,000	0.040345
Emissions in kg CO ₂ eq per gram fruits and vegetables: Average Mean GHG Emissions per kg / 1,000	0.002147

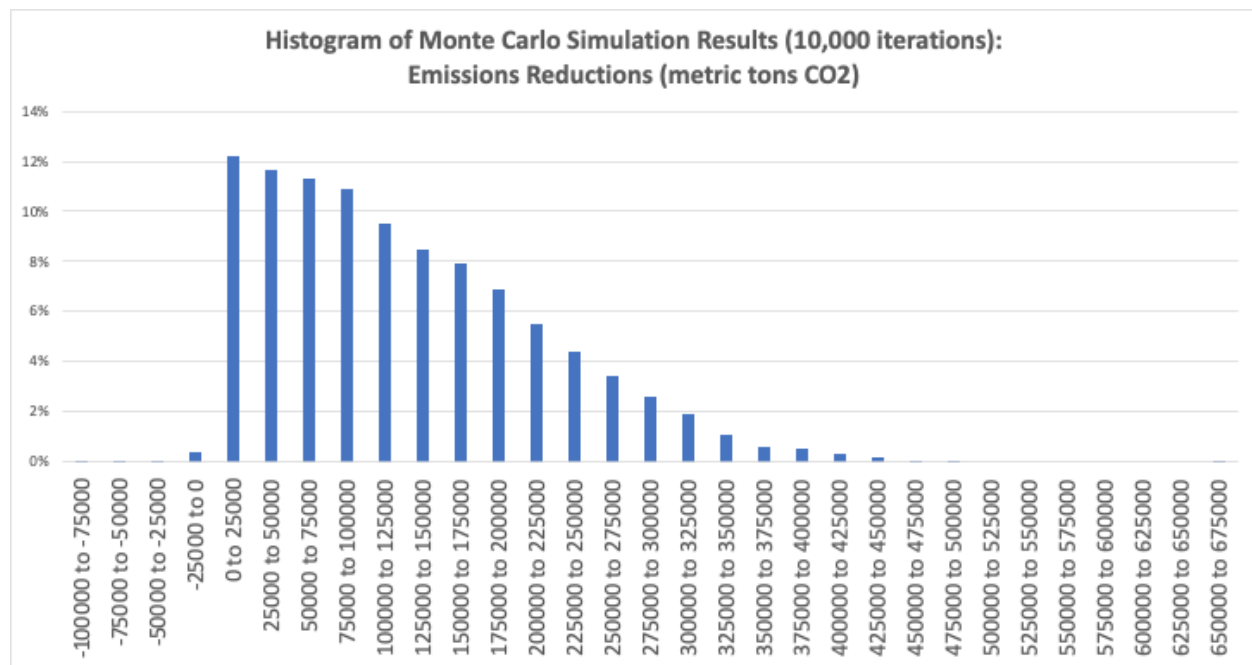
Table 13

Calculating Emissions	
Emissions (metric tons) per gram meat (average) (Emissions in kg CO ₂ eq per gram / 1,000)	0.000040345
Emissions (metric tons) per gram fruits and vegetables (average) (Emissions in kg CO ₂ eq per gram / 1,000)	1.7595E-06
Grams per serving meat (average)	155.925
Grams per serving fruits and vegetables (average)	84.91190476

Table 14

Monte Carlo Parameters	
Cost (25 Stores)	
1625000 (min.)	2500000 (max.)
Changes in Daily Servings	
0.2971 (mean)	0.1128 (standard dev.)
Participation (population)	
0 (min.)	375039 (max.)

Graph 1



FCEV Sharing Program

The last proposed program would be the FCEV sharing program, which mainly focuses on outsourcing the program to StratosFuel and providing FCEVs to the community members who need to commute.

The StratosShare program, launched in Riverside and San Bernardino counties in early 2019, provided affordable solutions in offering hydrogen-powered fuel cell electric vehicles (FCEVs) (U.S. Department of Energy, 2019). Except from FCEVs, through StratosFuel's collaboration with Toyota, the company developed an app to inform the users to find and use one of the 17 Toyota Mirai (\$49,500 per car) FCEVs and offer 24/7 customer service in guiding the customers to roadside services and other instructions on the road (U.S. Department of Energy, 2019). The APP will also update the real-time information regarding energy usage to ensure that the tank is full and provide customer service to those who have never been exposed to FCEVs before (U.S. Department of Energy, 2019).

EV prices still are a huge factor to EV readiness in urban communities. EVs owners are currently likely to be wealthier, more educated, and living in suburban areas (Pasha et al., 2016). However, many predicted that the price for EVs would be likely to decrease and EVs would be more affordable in the future. FCEV sharing program could be an educative tool for low-income community members to get exposure to FCEVs. The FCEV sharing project is a more progressive program compared to other alternatives such as building more charging stations and providing public transit vouchers to the low-income community members.

Costs of an FCEV Sharing Program

The total grant for this particular program amounted to \$684,421 (NPV: \$620,790), which mainly used for App development, hydrogen power station infrastructures, and FCEV purchasing, follow-up repairing and upgrading, and management costs (U.S. Department of Energy, 2019). The upfront costs may seem daunting, but in the long term, the costs would be less as more clean energy was utilized efficiently.

This FCEV sharing program provided an affordable solution to travel sustainably and at a zero-emission rate in disadvantaged communities, pairing with more high-capacity hydrogen power stations. The rate is as low as \$5 per hour without any membership, refueling, or insurance fees, customers could rent one Toyota Mirai within a VMT of 100 miles of Riverside (U.S. Department of Energy, 2019). Among all transportation solutions in Riverside and San Bernardino, this could be the cheapest and most available to commuters, which could potentially increase the usage and drive down more GHG emissions (U.S. Department of Energy, 2019).

Feasibility/Effectiveness of an FCEV Sharing Program

The total time to fuel up the energy from zero is within 5 minutes, and if the driver visits the StratosFuel site, there will be a 30-minute free rental time, serving as an incentive to maximize the utility of FCEVs and attract the registered users (U.S. Department of Energy, 2019). With these incentives offered to drivers, the StratosShare already has 500 pre-registered users, before its scheduled launch, which proved that this program has a potential to serve more people and save their money in purchasing a fossil-fueled vehicle (U.S. Department of Energy, 2019).

The company estimated that their fleet of FCEVs could reduce more than 3000 metric tons of harmful emissions within five years (U.S. Department of Energy, 2019). To be more specific, the cost-effectiveness to run this program is \$41.39 per ton over a five-year span. According to StratosFuel's CEO and cofounder, one FCEV could replace at least 9 to 13 regular cars powered by fossil fuels (U.S. Department of Energy, 2019). As the program expands to more jurisdictions, more would opt FCEVs over regular rental cars. By 2020, the company ramps up its Zero Impact Production (ZIP) facilities by 150% and increases more fueling station locations (U.S. Department of Energy, 2019). In the future, the company would tap into its user data and analyze the potential demand for more FCEVs and more fuel to adjust the company's FCEV and fuel supply (U.S. Department of Energy, 2019).

Table 15

Sign-up headcounts	≥ 500
Annual carbon emissions reduction potential (metric tons CO ₂)	30,000
Cost (present value)	\$0.68 million
Cost per ton	\$41.39
Net present cost	\$0.62 million

Equity and an FCEV Sharing Program

This EV sharing program is a non-emission alternative and also a demonstration to all California low-income communities that an EV could be available to them without excessive costs. For the parking space of FCEVs, the company also has designated options for community members. The drop-off sites usually include downtown parking garages, universities, and high-density residential areas (U.S. Department of Energy, 2019). To enable more convenient sites, StratosFuel partnered up with a variety of commercial parking companies, organizations, and communities.

Co-Benefits to the Community of an FCEV Sharing Program

According to a report published by the US Department of Energy, the combustion of fossil fuel energy is the main contributor of emission of GHGs, smog-forming compounds, particulate matter, and other air pollutants, and the leading causes of climate change (U.S. Department of Energy, 2016). The application of EVs can reduce the dependence on fossil fuels and the health threats as a result of climate change.

The global health impact of climate change is alarming. It is estimated that between 2030 and 2050, climate change will cause approximately 250,000 additional deaths per year, as a result of malnutrition, malaria, diarrhea, and heat stress (U.S. Department of Energy, 2016). The direct global damage costs are between \$2 billion and \$4 billion per year by 2030, which will disproportionately fall on low-income and vulnerable populations (U.S. Department of Energy, 2016). A study conducted by Rand Health, a non-profit research institution, found that “between 2005 and 2007, nearly 30,000 emergency-room visits and hospital admissions in California could have been avoided and a \$193 million cost could have been saved, if the national minimum air quality requirements had been met (U.S. Department of Energy, 2016).” Better air quality may not have a direct and obvious impact on health, however, in the long-term the health benefits would pay off.

If FCEVs were employed to solve local community’s needs to transport, then FCEVs would render a safe means of transportation for everyone (U.S. Department of Energy, 2014). This program will benefit more people if they have the necessary vehicles to land a new employment opportunity which requires more than one hour to commute.

Financing and Tax Incentives

To push forward the vision of sustainable development, California state regulations have been supportive in offering tax rebates to local governments, businesses, and the local community who purchased EVs and EVSEs for commercial and residential use (California Clean Vehicle Rebate Project, “Electric Vehicle Charging Overview”, 2018). For example, the Los Angeles Department of Water and Power (LADWP) offered different tax rebates ranging from \$750 to \$15000 to commercial purchases depending on the charger type. Residents acquiring plug-in EVs could receive \$750, regardless of the type and model. Besides tax benefits, California Electric Vehicle Infrastructure Project offered funds for further infrastructure development of EVSE and DC fast-charging on a first-come-first-serve basis (California Clean Vehicle Rebate Project, “Electric Vehicle Charging Overview”, 2018). However, tax benefits usually don’t have an immediate impact on the upfront payment when purchasing EVs, therefore, financing for neighboring residents to buy EVs still could seem unrealistic if they have to pay a lump sum of down payment first (U.S. Department of Energy, 2014).

The program’s funding company acknowledged that the key to success in the program is to secure an alliance between the provider and the local Clean Cities Coalition, Western Riverside County Clean Cities (WRCCC) (U.S. Department of Energy, 2019). USC could consider the possible community FCEV sharing project in its neighborhood communities and form an alliance that benefits the communities. The implementation could be determined by USC and the contractor for the details of this plan.

Criteria-Alternatives Matrices

Table 16 details the results of the primary criteria-alternatives analysis using the five standard criteria identified by the practicum team for inclusion of carbon offset options in a final carbon offset strategy recommendation. Definitions and ratings follow the conventions outlined in the *Methodology* section. As observed in *Table 16*, the healthy markets option appears to be the “best” carbon offset option at face value, given the total criteria score of “20”. Adhering to the group’s policy that any carbon offset option included in the final recommendation for the university’s carbon offset strategy must satisfy, at minimum, a total criteria score of “12”, the practicum group has determined not to recommend the residential weatherization or residential solarization options. Each of these proposed carbon offset programs failed to reach a minimum criteria score of “12”, weakened by high cost, low emissions reduction potential, and concerns regarding feasibility and equity. Of note, Duke University discontinued similar residential carbon offsets programs due to feasibility and verification concerns (DCOI).

Table 16

	Cost	Feasibility/ Efficiency	Co-Benefits to Community	Emissions Reduction Potential	Equity	Total
Residential Weatherization	1	2	4	1	2	10
Residential Solarization	1	2	4	1	2	10
Small business solar transition (400 businesses)	1	3	3	3	4	14
Healthy Markets	5	2	3	5	5	20
FCEV Sharing	5	4	3	3	3	18

Table 17 below details the results of the secondary criteria-alternatives analysis under the P.A.V.E.R. framework. Definitions and ratings follow the conventions outlined in the *Methodology* section. While the USC Presidential Working Group on Sustainability specified in initial interviews in August and September of 2020 that it is not entirely concerned with carbon

offset option satisfaction of each of the P.A.V.E.R. requirements, the practicum has included this analysis should the university decide at a later date to petition for recognition of its carbon offsets under California Health and Safety Code. If the university does decide to pursue state recognition of carbon offsets, the recommended carbon offset strategy would include only carbon offset options that achieve a rating of “1” under all five of the P.A.V.E.R. criteria. Any carbon offset option that exhibits a rating of “0” under any of the P.A.V.E.R. requirements will fail to be recognized as a legitimate carbon offset under California Health and Safety Code §38562(d)(1) and (d)(2).

Table 17

	Permanent	Additional	Verifiable	Enforceable	Real
Residential Weatherization	0	1	1	1	1
Residential Solarization	1	1	1	1	1
Small business solar transition (400 businesses)	1	1	1	1	1
Healthy Markets	1	1	0	1	1
FCEV Sharing	1	1	1	1	1

Feedback from Stakeholder and Expert Interviews

Throughout the process of research and analysis for this report, the practicum team was able to interact with a range of stakeholders and experts. This section will summarize the major takeaways from these interviews, including the community listening section.

This report was informed by interviews with members of USC faculty and staff, including some of those on the Presidential Working Group on Sustainability. Outside of the PWG, we met with faculty from the Keck School of Medicine of USC and faculty from the Price School of Public Policy. We also interviewed staff from around the University, such as from the USC Good Neighbors Campaign, the USC Sustainability office, and USC’s office of Government and Civic Engagement. These interviews spanned a wide range of topics and, accordingly, provided a wide range of thematic takeaways. First, and maybe most obviously, is that USC should engage the Good Neighbors Campaign in any project it takes. As USC already has an office dedicated to community outreach, it makes sense for such projects to work closely with this office. Second, through the Good Neighbors Campaign, it may be an attractive idea to use RFPs to not only generate project ideas but also initiate strategic partnerships with stakeholders. The Good

Neighbors Campaigns already accepts RFPs for grant money for community projects, and USC may identify possibilities for partnerships, student involvement, research, and carbon offsetting all while engaging the community in this fashion. Third, the University should use and promote any project as an emerging research opportunity, as well. There are numerous professors and researchers on campus whose work relates to this issue, and it may be prudent to combine efforts at carbon offsets with opportunities for faculty and students to get involved. Relatedly, students are likely to want to express their views on projects related to sustainability and to the community, and students also are a valuable resource for research. There is plenty of room for student involvement in any project going forward, and it may make it more attractive to a wider audience, too. Fourth, USC can and should look into partnerships with other institutions in the surrounding communities, such as hospitals. Other nonprofits, too, would make strong partners for the University in community-related programs. Finally, we might also consider interventions that are aimed at the off-campus but proximate student population. This population is geographically mixed with non-USC communities in the surrounding neighborhoods and may be an easy group to reach with an off-campus initiative.

In addition to interviews with USC personnel, we met with experts from other parts of the country, such as from the American Council for an Energy-Efficient Economy and from the Duke University Office of Sustainability. Overall, the following themes emerged from the interviews with these experts. First and foremost, the University should decide how it views recognition of carbon offsets, and whether this would be an internal project or something it would like to publicly verify. The difference between each option is that, typically, carbon offsets use or develop carbon offsets protocols in order to be verifiable from the viewpoint of a third party. Essentially, there are steps an entity can take to receive peer review for its offsets, and doing so would benefit greatly from the using existing or developing new carbon offsets protocols. Second, behavioral change interventions are an interesting alternative to structural or systems improvements, which are typical of carbon offsets projects. In theory, according to many of those with whom we spoke, carbon offsets due to investments in behavioral change are a viable option and an idea that may gain a lot of interest. As such, we did not limit this analysis to alternatives with existing protocols, and instead thought more broadly about what is possible to reduce carbon emissions. The final takeaway is that the University is an academic leader whose strengths are in research and education. Therefore, USC should consider the scope of its own role in the project as that of an educator of the public, and utilize the tools it has at its disposal. Therefore, the expertise to conduct the project, itself, might belong to another entity with whom the University could work while USC focuses on education (including of the community) and research.

Finally, on April 13, 2021 we met with a small group of community members from the areas surrounding the University Park Campus and the Health Sciences Campus. These individuals included longtime residents and neighbors of the University, local government leaders, people engaged in nonprofit organizations, and others who are involved in civic work. The purpose of

this listening session was to introduce members of the surrounding communities to our project idea and ask for their input on things to consider, as well as their thoughts on what we had at that point. The session was very constructive and resulted in a number of conclusions we were able to take away. First and most importantly, the community members by and large expressed their desire for blended efforts between the University and the community. Some members expressed their perceptions of the University-community relationship as having separated calcified over time, and stated a wish to see the University actively engage with its neighbors in a more normalized and consistent manner. That is, rather than engaging through limited channels, they want to feel a greater sense of community and see the University as partners and neighbors, not just a proximate entity. Second, there was a very strong preference in the session for projects that emphasize cleaner living space, such as through parks and greenery. This related to the desire that community members expressed for community-wide interventions rather than those that benefit only individual households. Community members also similarly liked the idea of healthy food grocery stores, as those are difficult to come by in their neighborhoods. Therefore, in addition to mutual engagement between the community and the University, community members want investments that improve that community as a whole. Third, the panel wants to see the University lead by example. Rather than tell residents and businesses around its campuses how to change their behaviors, they want to see USC do its part, too. This led to a productive conversation where we discussed the efforts the University has undertaken to do just that, however it highlights that community engagement presents a communication challenge; even though the University has been making structural and systems changes to promote sustainability, local leaders were not necessarily aware of those actions. This links back to the emphasis on community-University blending and communication. Any project the University wants to pursue requires conscious public relations efforts to build support and trust with the community.

Finally, there are some existing and upcoming opportunities for engagement with the community that received mention in the discussion. First, as the Olympic Games are coming to Los Angeles soon, USC and the community could use that as an occasion to showcase work on sustainability that they have done together. To do this, USC should think about how to engage and elevate community voices now and get an early start on any initiatives together. To do so, panelists recommended inviting community members to regular discussions on this topic with the University and then to later designate a team of community leaders that work even more closely with USC's team. Second, because of the proximity of students and non-student community members, panelists would love to see efforts to foster and improve relationships between the two. Third, the panelists recommended that the University not tie any investments to the strict boundaries of the Good Neighbors Campaign, but rather to think of the wider impact and ways that the entire communities, and not just portions, can benefit. Overall, the listening session highlighted the importance of conscientious relationship-building between the University and the entire surrounding community.

Recommendations

Given the tendency of multicriteria decision analysis to result in recommendations that ignore certain stakeholder groups (Kiker et. al., 2005), the practicum team recommends the pursuit of several carbon offset options, with none having priority over the others. Instead, the team's analysis indicates that further exploration of a carbon offset strategy that encompasses a suite of carbon offset options is worthwhile.

Additionally, due to the risk associated with offsets, it may be wise for the University to diversify its portfolio of projects. The University could both purchase offsets off of the international market and work to develop a number of projects locally. Some of the faculty this project engaged also mentioned that projects could be multi-faceted and may be designed as a “package” of interventions. For example, projects that reduce carbon emissions by encouraging micro-mobility in South and East Los Angeles might also involve traffic calming efforts, could incorporate greening of streets and sidewalks, and necessitate partnerships with local institutions, such as city planning. Similarly, projects that seek to support energy burdened households could collaborate with green workforce initiatives in the region, community health centers, and other groups, for example.

A clear finding of the research was that the University has a breadth of interdisciplinary expertise related to climate change and carbon emissions reduction and mitigation. Whether in its engineering department for modeling energy reductions and developing new technologies or in faculty and students with interests in mitigation pollution from transportation, this project has determined that there is potential for the PWG to work with a wide range of on-campus groups. It will also be important for the PWG to collaborate with USC's fundraising and development arms to develop offset projects with high-impact potential. Investment in offsets could furthermore be part of USC's interdisciplinary academic research agenda, by providing opportunities to really push for sustainability and climate change mitigation in local communities.

Notably, the community stakeholders from the listening session mentioned not wanting to be “guinea pigs” for new research and instead wanting to be part of the process that drives project development. Thus, it will be important to engage in community participatory planning and research when developing ideas. One interviewee mentioned that USC's offset agenda could consider the “asks” that community leaders or stakeholders have already made in the past to determine whether they could have a carbon reduction aspect. For example, community stakeholders have previously expressed interest in having access to USC's shuttle service, having access to more green space, and having concerns with USC's land use and gentrification forces. This perspective demonstrates how community groups can have useful feedback about project types that would be considered acceptable in the development of a novel carbon offsets program.

This research recommends developing intentional, long-term relationships with community stakeholders and potential institutional partners for the development of projects. This may include nearby colleges and universities, internal organizations, such as the Keck School of Medicine, or external bodies, such as local government agencies and other environmental nonprofits with similar missions. A problem that may emerge with local partnership projects is the issue of double counting. Some stakeholders we interviewed mentioned that collaboration with electric utility companies or organizations with similar missions may require clear agreements for determining which party will claim offset credits. However, this issue may not be as important if the University's overall motivation is to work with others to support the transition of local communities to a low-carbon future. Namely, climate change research warns that businesses, institutions, and local governments must act quickly to curb carbon emissions. For further expertise on measurement, verification, and credibility of projects, we recommend leveraging the potential for partnerships with verifier groups, such as the Offset Network, a group of higher education institutions that seek to make offsetting accessible, scalable, and innovative for universities, particularly by utilizing peer verification of projects.

Lastly, in developing an offset program, it is important for the University to consider that it might be better to think of certain projects as falling under near- or long-term strategies. Certain projects may be easier to implement than others and support the University in quickly reducing carbon. Other projects, including those that reduce Scope 1, 2, and 3 emissions, may be more expensive and require longer time frames (Lee and Mayer, 2020).

Directions for Future Research and Work

During the practicum project, we focused on the projects that merited initial attention, but there are certainly some other perspectives that need to be studied, especially in the post-COVID era. Future studies could potentially focus on the following concerns: ways to shorten the offset learning curve for the university and its peers, ways to complement the ongoing efforts by the City and County (i.e., such as LA's Green New Deal), an evaluation of how to integrate USC's offset strategy with the post-Covid recovery lens of improving job opportunities (i.e., green economy), improving local educational opportunities, or promoting healthy living and equitable recovery.

It is important for future work to determine how USC will develop different facets of its offset strategy. It will need to decide whether to engage in third-party or peer verification offset projects (See a snapshot of certifier information in *Appendix D*). Future research could also explore the potential for co-development of interdisciplinary offset protocols by university faculty and students.

Additionally, future research could further explore root source reduction of Scope 1, 2, and 3 emissions. For example, some interviewees brought up new ideas that did not fit the scope of our project (because they targeted students living off campus or faculty), but could be implemented on-campus. For example, one interviewee brought up elective programs to offset intensive airline travel for the USC community.

There are also various aspects of each proposed project in this report that need further investigation. In this report, we did not focus on the potential for building a solar farm or array that benefits more than a single-household, which could have a broader impact on many people living near the vicinity of USC. For residential weatherization and solarization, we noted that multi-unit rental or low-income housing might benefit more from interventions, but more research is needed to understand how to make these projects feasible and affordable. For the healthy food market option, there will need to be more evaluation of a dietary change's impact on carbon reductions and there would need to be exploration of program evaluation for these types of healthy markets. For transportation, FCEVs are a fairly new concept, and more research is needed to determine the demand for FCEVs and EV charging stations and other types of EVs or travel modes that exist in USC's local communities .

Conclusion

Throughout this project we have worked closely with the USC PWG to develop a set of project scenarios that were developed and discussed with different faculty, experts, and community stakeholders, while attempting to provide the quantified carbon benefits for environmental upgrades to nearby individuals, households, and small businesses of the communities in South and East Los Angeles. Due to limited empirical research, the team notes the number of potential co-benefits to these communities stemming from the emerging literature on carbon offsetting, but quantification and valuation of co-benefits merits further attention to inform decision making.

The team has mapped out several alternatives and projected costs and emissions reductions from various applications of these alternatives. The team has also developed evaluative criteria to weigh these alternatives against the baseline carbon offset of purchases on an international market. Recommendations are made regarding potential programs the university can employ to meet carbon neutrality goals. There are concerns with implementation of each project alternative, as most of the project proposals feature high costs and require ongoing efforts over a long period of time. With further research and potential beta testing of certain carbon offset proposals presented in this report, the university can move toward carbon neutrality while leading on a novel hyper-local carbon offset portfolio.

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Appendices

Appendix A) Relative Risks of Carbon Offset Project Examples

Project Type	Risk Level	Co-Benefits	Harms
CO2	Low		
Methane Destruction	Low	May reduce odor from landfills	Could align with the perpetuation of the coal industry, deferring climate change goals.
N2O Avoidance from Nitric Acid Production	Low		Could align with the synthetic fertilizer industry.
N2O Destruction in Adipic Acid Production	Low		
Destruction of Ozone Depleting Substances (ODS)	Low		Destruction of ODS helps to accelerate recovery of the stratospheric ozone.
Methane Capture and Utilization for Energy	Medium	Reduces air pollution if used to displace coal consumption	Could align with the perpetuation of the coal industry, thus deferring climate change goals.
Methane Emission Avoidance	Medium	Help reduces food wasted and promotes organic farming	
Energy Distribution	Medium	Significant air quality benefits by displacing inefficient domestic combustion methods; connect isolated households to the electrical grid.	
Energy Efficiency for household demand	Medium	Cost-savings for end-users, meaningful public health improvements	
PFC & SF6 Avoidance and Use	Medium		
Renewable Energy, Small Scale (Under 15 MW)	Medium	Reduced air pollution if fossil fuel consumption is displaced.	Increased air pollution if fossil fuel controls are excluded.

*Adapted from content provided by Offsetguide.org, a project developed by the GHG Management Institute and the Stockholm Environment Institute (2021).

Appendix B) List of USC and External Interviewees

Name	Title	Topic
Dr. Marlon Boarnet	Chair of the Urban Planning & Spatial Analysis Department	Transportation offset alternatives
Dr. Carolina Castillo	Executive Director of Development for USC Civic Engagement	Integrating community-serving projects with the food insecurity and sustainability priorities of the Good Neighbors Campaign
Dr. Lavonna Lewis	Associate Dean of Diversity, Equity, and Inclusion	Health implications of carbon offset projects and localized partnerships
Dr. Kayla de la Haye	Assistant Professor of Preventive Medicine	Food insecurity, healthy markets, and dietary implications of offsetting projects
Dr. Wandi Bruine de Bruin	Provost Professor of Public Policy, Psychology, and Behavioral Science	Behavioral interventions for diet and climate change mitigation
Matthew Arsenault, MPP	Duke Carbon Offsets Initiative Staff	Developing offset projects from a university standpoint
Dr. Reuven Sussman (ACEEE)	Director, Behavior & Human Dimensions Program	Behavioral change interventions for energy efficiency and waste
Zelinda Welch, MBA	Facilities Manager at USC	Solar and implementation considerations
David Galaviz (USC External Relations Team)	Associate VP, Government and Community Relations	Community partnerships and perspectives
Community Leaders	Lincoln Heights Neighborhood Council, South LA Neighborhood Council, Tree People	Feedback on proposed options and ideas for future offset strategy

Appendix C) List of Independent Certifiers

Name	Requires 3rd Party Verification	Separates Verification and Approval Processes	Project Types	Link
Verified Carbon Standard (Verra)	Yes	No	All sectors except new HFC Refrigerants	https://verra.org/project/vcs-program/
Clean Development Mechanism	Yes	Yes	All sectors except Reducing Emissions from Deforestation and Degradation, new HFC Refrigerants and nuclear	https://cdm.unfccc.int/about/index.html
Gold Standard	Yes	Yes	Energy Efficiency and Land Use	https://www.goldstandard.org/
Plan Vivo	No	No	Land Use, Land-Use Change, Forestry	https://www.planvivo.org/
Climate Action Reserve	Yes	Yes	All sectors	https://www.climateactionreserve.org/how/projects/
Offset Network	No	Yes	University-developed Offsets	https://offsetnetwork.secondnature.org/

*Content for chart adapted from Lee & Mayer (2020)